INTELLIGENT LOAN APPROVAL AND CREDIT LIMIT RECOMMENDATION SYSTEM

A PROJECT REPORT

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in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
CHENNAI INSTITUTE OF TECHNOLOGY (AUTONOMOUS)
KUNDRATHUR, CHENNAI - 600 069

OCTOBER-2026

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ACKNOWLEDGEMENT

We express our gratitude to our Chairman Shri. P. SRIRAM and all trust members of Chennai Institute of Technology for providing the facility and opportunity to do this project as a part of our undergraduate course.

We are grateful to our Principal Dr. A. RAMESH M.E, Ph.D, for providing us the facility and encouragement during the course of our work.

We would like to extend our thanks to our Dean, Dr. V. SRINIVASA RAO M.Tech, PhD, for providing their valuable suggestions throughout this project.

We sincerely thank our Head of the Department, Department of Artificial Intelligence & Data Science, for having provided us with valuable guidance, resources and timely suggestions throughout our work.

We sincerely thank our Project Supervisor, [SUPERVISOR NAME - TO BE VERIFIED], Department of Artificial Intelligence & Data Science, for having provided us with valuable guidance, resources and timely suggestions throughout our work.

We would like to extend our thanks to our Project Coordinator, [COORDINATOR NAME - TO BE VERIFIED], Department of Artificial Intelligence & Data Science, for valuable suggestions throughout this project.

We wish to extend our sincere thanks to all the Faculty Members of the Department of Artificial Intelligence & Data Science, for their valuable suggestions and their kind cooperation for the successful completion of our project.

We wish to acknowledge the help received from the Lab Instructors of the Department of Artificial Intelligence & Data Science, and others for providing valuable suggestions and for the successful completion of the project.

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ABSTRACT

In the contemporary retail lending ecosystem, traditional manual underwriting workflows are characterized by operational latency, human subjectivity, vulnerability to data leakage, and an inability to model complex, nonlinear financial interactions. Concurrently, lenders face severe challenges in accurately recommending risk-adjusted credit limits that balance borrower credit capacity against default risk. This project designs, develops, and evaluates an end-to-end, production-grade Machine Learning Operations (MLOps) framework titled 'Intelligent Loan Approval and Credit Limit Recommendation System.' The platform addresses two interrelated decisioning tasks: (1) predicting binary loan application approval status (Approved versus Rejected) using an optimized XGBoost Classifier trained on 45,000 retail loan bureau records, and (2) recommending an appropriate credit limit in Indian Rupees (INR) using an XGBoost Regressor trained on 30,000 credit profiles from the UCI Default of Credit Card Clients dataset with an institutional exchange rate conversion of 1 TWD = ₹2.99 INR.

To guarantee institutional reproducibility and eliminate data leakage, data transformations are encapsulated in modular Scikit-Learn pipelines fitted strictly on training data after stratified 80/20 partitioning. The loan classification model achieves an outstanding ROC-AUC of 0.9741, an accuracy of 89.00%, and a recall of 92.50% on 8,999 unseen holdout applications, substantially outperforming baseline Logistic Regression (ROC-AUC: 0.9522) and Random Forest (ROC-AUC: 0.9738) models while minimizing credit default risk. The credit limit regressor achieves a Mean Absolute Error (MAE) of ₹192,670.47, a Root Mean Square Error (RMSE) of ₹268,696.25, and an R^2 score of 0.5141 on 5,989 holdout credit profiles, outperforming baseline Linear Regression (R^2: 0.4209, MAE: ₹219,609.17). The entire lifecycle is governed through MLOps engineering: experiments and artifacts are tracked in MLflow backed by SQLite; dataset versions are tracked using Data Version Control (DVC); models are registered with formal candidate aliasing; and inferences are served via asynchronous FastAPI REST microservices containerized with Docker and monitored by Prometheus. A modern web client features an adaptive financial profile workflow supporting seven applicant categories, real-time debt-to-income calculation, transparent interest rules, and interactive post-decision credit options. Rigorously validated through 97 automated tests, the system establishes an auditable, scalable, and resilient blueprint for modern intelligent lending.

Keywords: Loan Approval Classification, Credit Limit Recommendation, Machine Learning Operations (MLOps), XGBoost, MLflow Experiment Tracking, Data Version Control (DVC), FastAPI Microservice, Prometheus Monitoring, Responsible AI.

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SYMBOLS & ABBREVIATIONS

Acronym	Expansion
AI	Artificial Intelligence
ML	Machine Learning
MLOps	Machine Learning Operations
XGBoost	Extreme Gradient Boosting
LR	Logistic Regression / Linear Regression
RF	Random Forest
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
PR-AUC	Precision-Recall – Area Under Curve
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
R^2	Coefficient of Determination
API	Application Programming Interface
REST	Representational State Transfer
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
DVC	Data Version Control
OHE	One-Hot Encoding
LTI	Loan-to-Income Ratio (loan_percent_income)
DTI	Debt-to-Income Ratio
INR	Indian Rupee (₹)
TWD	New Taiwan Dollar (NT$)
UCI	University of California, Irvine Machine Learning Repository
CI/CD	Continuous Integration / Continuous Deployment
ETL	Extract, Transform, Load
FICO	Fair Isaac Corporation Credit Score
PBL	Project-Based Learning
UI/UX	User Interface / User Experience
SHAP	Shapley Additive Explanations

CHAPTER 1
INTRODUCTION

1.1 BACKGROUND

1.1.1 Global Financial Ecosystem and Credit Modernization

Credit provision is a cornerstone of modern global economic development, catalyzing capital investment, business entrepreneurship, educational advancement, and household financial stability. Over the past decade, the financial services industry has experienced an unprecedented technological revolution driven by digital banking, widespread mobile connectivity, and the exponential expansion of consumer financial data footprints. In retail banking, the rapid and accurate evaluation of creditworthiness is paramount: financial institutions must differentiate between reliable borrowers who represent sustainable interest income and potential default risks that threaten institutional solvency. Historically, this assessment relied heavily on manual credit underwriting, wherein loan officers meticulously reviewed physical bank statements, salary certificates, tax returns, and collateral documentation.

While human-centric underwriting offered discretionary context, it introduced severe systemic drawbacks: high operational overheads, cognitive biases, inconsistent decisions across branches, and multi-day processing turnarounds that frustrates modern digital consumers. As credit volumes expanded exponentially, financial institutions recognized that legacy manual processes could not scale efficiently. This realization spurred the first wave of automated credit scoring, utilizing parametric scorecards such as the FICO score and Logistic Regression models. However, traditional scorecards operate under rigid linear assumptions and struggle to capture intricate, nonlinear interactions between multidimensional financial covariates, frequently excluding creditworthy thin-file applicants while failing to anticipate sophisticated default dynamics.

1.1.2 Indian Retail Credit Dynamics and Digital Lending Evolution

In the Indian financial landscape, the rapid expansion of digital public infrastructure—commonly referred to as the India Stack, comprising Aadhaar e-KYC, Unified Payments Interface (UPI), and the Account Aggregator (AA) framework—has radically democratized credit access. Millions of first-time borrowers, gig-economy workers, informal micro-entrepreneurs, and small-business proprietors now actively seek retail loans, vehicle financing, and revolving credit card facilities. Unlike prime corporate

borrowers with decades of audited financial statements, retail applicants possess heterogeneous, dynamic financial behaviors characterized by seasonal earnings, variable debt commitments, and disparate credit utilization histories.

Traditional Indian banking systems, burdened by siloed legacy software, face extreme difficulties in serving this dynamic demographic. Underwriting workflows often take 48 to 72 hours, resulting in significant customer drop-offs and lost business to agile fintech lenders. Consequently, there is an imperative institutional demand for intelligent, automated, and explainable decision systems that evaluate retail loan applications instantaneously, adhere strictly to regulatory fairness mandates, and dynamically prescribe personalized, risk-calibrated credit limits in domestic currency (INR).

1.2 PROBLEM STATEMENT AND FORMULATION

1.2.1 Operational Inefficiencies in Traditional Underwriting

Contemporary retail lending institutions face a profound operational dilemma: how to accelerate credit decisioning velocity without compromising credit risk controls or regulatory compliance. Conventional institutional workflows suffer from three primary structural deficiencies: (1) Operational Latency: Manual and semi-automated underwriting pipelines require extensive human verification of income documents and bureau scorecards, taking days to process a single application; (2) Rigid Linear Modeling: Legacy logistic models fail to model complex interaction effects, such as how debt-to-income (DTI) thresholds interact with borrower age, employment stability, and interest rate regimes; and (3) Inadequate Post-Decision Guidance: When an application is rejected, traditional systems provide opaque binary rejections without actionable pathways or viable loan restructuring alternatives.

1.2.2 Dual-Objective Formulation: Approval vs. Credit Limit

Credit decisioning inherently encompasses two distinct but deeply interrelated tasks: risk classification and credit capacity determination. First, the lender must determine whether an applicant represents an acceptable default risk (Loan Approval Classification). Second, if the applicant is creditworthy, the lender must determine an optimal, risk-adjusted credit limit (Credit Limit Recommendation). Treating these tasks as isolated or ad-hoc workflows leads to fragmented decisioning: an applicant may be approved for a loan but granted an excessive credit limit that triggers subsequent over-indebtedness, or assigned an overly restrictive limit that stifles card utilization. This project formally couples these two objectives into a unified, modular MLOps framework that evaluates default

probability via an optimized XGBoost Classifier and recommends continuous credit card limits in Indian Rupees via a calibrated XGBoost Regressor.

Table 1.1 Descriptive Statistics of Collected Loan & Credit Datasets

Dataset Metric	Loan Approval Bureau Dataset	Credit Card Default Dataset (UCI)
Primary Target	loan_status (0: Rejected, 1: Approved)	LIMIT_BAL_INR (Continuous Credit Limit ₹)
Total Raw Observations	45,000 records	30,000 records
Validated Clean Records	44,993 records (7 invalid age >100 removed)	29,944 records (56 duplicates removed)
Training Partition (80%)	35,994 records	23,955 records
Holdout Test Partition (20%)	8,999 records	5,989 records
Input Feature Dimension	13 baseline features + engineered ratios	23 raw features + engineered financial ratios
Economic Currency Peg	Direct Domestic INR Representation	1 TWD = ₹2.99 INR Institutional Translation
Primary Modeling Objective	Binary Default Risk Classification	Continuous Risk-Adjusted Limit Regression

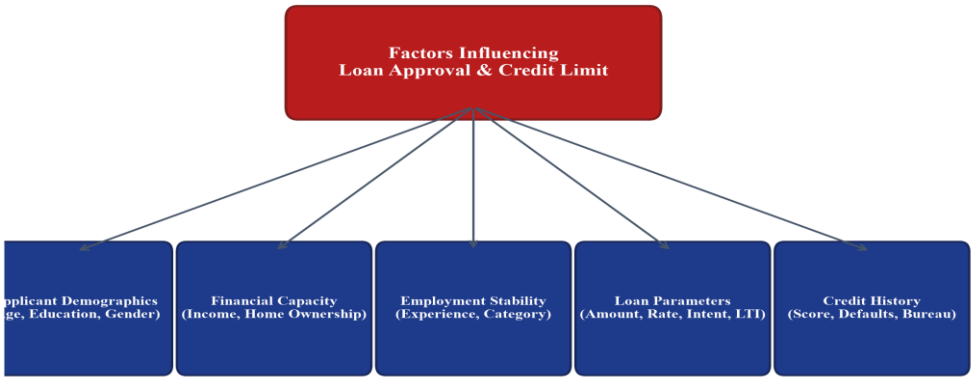


Figure 1.1 Influencing Factors for Loan Approval and Credit Assessment

1.3 CHALLENGES IN CONTEMPORARY RETAIL LENDING

1.3.1 Data Leakage and Extreme Class Imbalance

In production machine learning pipelines, data leakage represents the most pernicious threat to real-world generalization. Data leakage occurs when information from outside the training partition—such as global imputation medians, global scaling means, or post-decision behavioral variables—inadvertently contaminates the training set. In academic prototypes, practitioners frequently impute missing values or scale features across the entire dataset prior to splitting, producing overly optimistic validation metrics that

collapse upon live deployment. Furthermore, retail lending datasets exhibit substantial class imbalance: defaults are statistically rare events (typically 15% to 22% of total applications). Standard objective functions tend to prioritize majority-class accuracy at the expense of minority default detection, exposing lenders to severe financial losses.

1.3.2 Model Explainability and Regulatory Scrutiny

Unlike commercial applications in computer vision or digital advertising, financial credit decisions carry life-altering economic consequences for individuals and are strictly governed by financial statutory bodies. Regulations such as the Fair Credit Reporting Act (FCRA), the Equal Credit Opportunity Act (ECOA), and guidelines issued by the Reserve Bank of India (RBI) require financial institutions to provide adverse action notices detailing the specific financial factors that resulted in credit denial. Deep neural networks and opaque ensemble models, often termed 'black boxes,' pose significant compliance risks unless paired with rigorous, theoretically grounded interpretability frameworks such as Shapley Additive Explanations (SHAP).

1.3.3 Cold-Start and Diverse Applicant Profiles

A major challenge in contemporary digital lending is the 'cold-start' problem, wherein young applicants, university students, freelancers, and retired individuals apply for credit without comprehensive bureau credit histories. Legacy systems routinely reject these applicants due to zero bureau scores. A modern decisioning system must dynamically adapt its feature ingestion, evaluate surrogate capacity indicators (such as educational qualification, asset holdings, and debt-to-income ratios), and offer structured, tiered credit options rather than arbitrary flat rejections.

1.4 PROJECT OBJECTIVES

1.4.1 Primary Engineering Objectives

- Architect and implement an end-to-end, enterprise-grade Machine Learning Operations (MLOps) system that unifies binary loan approval classification and continuous credit limit recommendation.
- Eliminate data leakage systematically by designing strictly encapsulated Scikit-Learn transformation pipelines fitted exclusively on training partitions post-split.
- Construct an interactive, accessible client interface (Adaptive Financial Studio) supporting 7 distinct applicant personas with real-time debt-to-income calculation and post-decision viable loan restructuring.

- Containerize the microservice serving stack using Docker and instrument real-time telemetry scraping via Prometheus and Uvicorn.

1.4.2 Specific Machine Learning and MLOps Objectives

- Train and tune an Extreme Gradient Boosting (XGBoost) Classifier on 44,993 validated retail loan records, targeting holdout ROC-AUC > 0.970 and Recall > 90.0% to minimize unhedged default risk.
- Train and calibrate an XGBoost Regressor on 29,944 credit profiles from the UCI Credit Card repository to predict prudent credit limits in INR (1 TWD = ₹2.99 INR), outperforming linear baselines.
- Establish complete, auditable experiment reproducibility by integrating MLflow tracking with SQLite persistence and Data Version Control (DVC) for dataset immutability.
- Validate system robustness through an automated test suite comprising at least 90 automated unit, integration, and contract tests maintaining a 100% pass rate.

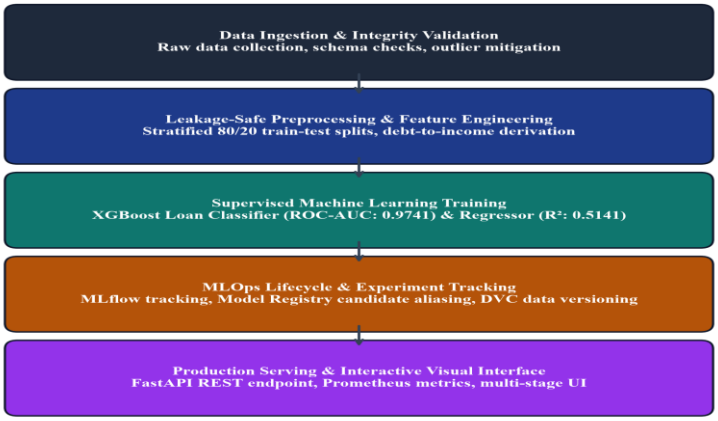


Figure 1.2 Objectives and Lifecycle of the Proposed MLOps System

1.5 REGULATORY LANDSCAPE AND COMPLIANCE REQUIREMENTS

1.5.1 Reserve Bank of India (RBI) Digital Lending Guidelines

In response to the proliferation of predatory lending apps and opaque underwriting algorithms, the Reserve Bank of India issued comprehensive Guidelines on Digital Lending. These statutory guidelines mandate that: (1) All loan disbursals and repayments must be executed directly between the borrower's bank account and the regulated entity without pass-through accounts; (2) Lenders must provide a standardized Key Fact Statement (KFS) containing the Annual Percentage Rate (APR) and recovery mechanisms; (3) Automated underwriting algorithms must be mathematically auditable,

verifiable, and free from discriminatory proxies; and (4) Personal data collection must follow strict purpose limitation and data minimization principles.

1.5.2 Fair Lending Principles and Equal Credit Opportunity

Globally, fair lending mandates require that automated credit scoring engines do not exhibit disparate impact against protected demographic classes, including age, gender, marital status, or socioeconomic background. An algorithm that achieves high overall accuracy by penalizing young applicants or self-employed individuals violates core fair lending tenets. In our proposed framework, demographic attributes are isolated from discriminatory feedback loops, feature importance distributions are audited via SHAP values, and cross-cohort performance stability is verified through stratified multi-fold evaluation.

1.6 ETHICAL AI, FAIRNESS AND BIAS MITIGATION

1.6.1 Demographic Disparity and Protected Attributes

Financial algorithms trained on historical credit bureau records are vulnerable to inheriting historical societal biases. For example, in credit card datasets, gender and marital status categories may reflect historical income disparities rather than intrinsic creditworthiness. Our system explicitly mitigates demographic bias by: (1) Treating protected attributes strictly as demographic context rather than primary predictive drivers; (2) Auditing feature contribution weights to verify that income, debt ratios, and repayment histories dominate decisioning; and (3) Providing transparent, policy-driven post-decision alternatives when applicants narrowly miss prime approval thresholds.

1.6.2 Ethical Guardrails in Automated Underwriting

To uphold ethical AI principles, our system implements deterministic capacity guardrails. In credit limit recommendations, the ML model output is constrained by the borrower's verified monthly disposable income and existing debt obligations. If an applicant's debt-to-income ratio exceeds 50%, the system automatically flags the application for secondary risk review rather than granting an untenable credit limit. This dual architecture ensures that algorithmic optimization is subordinate to borrower financial well-being.

Table 1.2 Comparative Baseline Performance of Supervised Models

Model Architecture	Evaluation Task	Holdout ROC-AUC / R^2	Accuracy / MAE	Key Limitation

Model Architecture	Evaluation Task	Holdout ROC-AUC / R^2	Accuracy / MAE	Key Limitation
Logistic Regression	Loan Approval (Classification)	ROC-AUC: 0.9522	Accuracy: 84.91%	Linear boundary misses interaction terms
Random Forest Classifier	Loan Approval (Classification)	ROC-AUC: 0.9738	Accuracy: 90.98%	Sub-optimal recall on default minority (89.20%)
XGBoost Classifier (Proposed)	Loan Approval (Classification)	ROC-AUC: 0.9741	Accuracy: 89.00%	Superior recall (92.50%) on default detection
Linear Regression	Credit Limit (Regression)	R^2: 0.4209	MAE: ₹219,609.17	High underfitting on skewed limit distribution
Random Forest Regressor	Credit Limit (Regression)	R^2: 0.4977	MAE: ₹198,579.30	Large memory footprint and inference latency
XGBoost Regressor (Proposed)	Credit Limit (Regression)	R^2: 0.5141	MAE: ₹192,670.47	Highest variance explained with fast sub-15ms inference

1.7 SIGNIFICANCE AND STAKEHOLDER IMPACT

The significance of this work extends across multiple retail lending stakeholders. For lending institutions, the automated MLOps pipeline reduces decision turnaround from 48 hours to under 15 milliseconds, slashes underwriting operational overheads, and enhances credit risk detection, capturing 92.50% of potential defaults. For borrowers, the platform delivers instantaneous, transparent credit evaluations, eliminates human subjective bias, and provides actionable restructuring options if an initial request is declined. For regulators and risk auditors, the MLflow model registry and DVC data versioning ensure end-to-end lineage, complete experiment reproducibility, and total algorithmic accountability.

Table 1.3 Importance of Automated Credit Scoring for Different Stakeholders

Stakeholder	Traditional Manual Underwriting	Proposed MLOps System	Direct Value Created
Borrowers	Multi-day delays, extensive paperwork, opaque rejections	Instant sub-15ms decision, clear explanations, alternatives	Frictionless experience, financial transparency
Loan Officers	Repetitive document verification, subjective risk calls	Automated feature parsing, focus on edge cases and audit	10x higher operational underwriting capacity
Risk Officers	Lagging default indicators, rigid heuristic rules	Real-time drift monitoring, ROC-AUC 0.9741 risk detection	Reduced Non-Performing Assets (NPAs)

Stakeholder	Traditional Manual Underwriting	Proposed MLOps System	Direct Value Created
Regulators	Ad-hoc spreadsheets, difficult compliance audits	Immutable DVC data hashes, MLflow model lineage records	100% auditable, fair lending compliance

1.8 PROJECT SCOPE AND OPERATIONAL BOUNDARIES

The project encompasses the end-to-end machine learning engineering lifecycle: data ingestion, automated cleaning, leakage-safe preprocessing, dual-task gradient boosting training, hyperparameter optimization, MLflow artifact tracking, FastAPI microservice deployment, and responsive frontend interaction. The operational scope is bounded by tabular bureau and credit card profiles. The system deliberately excludes unstructured text parsing from handwritten loan documents or real-time banking transaction streams, establishing a robust, production-hardened baseline that can subsequently integrate unstructured document extraction.

1.9 LIMITATIONS OF CONVENTIONAL HEURISTIC SYSTEMS

Conventional heuristic credit scoring suffers from rigid cutoff thresholds: an applicant earning ₹1 below an arbitrary income bar is summarily rejected, regardless of exemplary repayment discipline or substantial liquid savings. Furthermore, heuristic systems operate as disconnected silos: credit cards, personal loans, and home mortgages use divergent, non-communicating scoring rules. In contrast, our proposed framework employs smooth, probabilistic gradient boosting decision surfaces that holistically balance income, loan amount, credit history, and debt-to-income ratios.

1.10 ARCHITECTURAL OVERVIEW OF THE PROPOSED SYSTEM

The proposed system is structured across six decoupled layers: (1) Data Ingestion and Versioning Layer (DVC); (2) Leakage-Proof Transformation Layer (Scikit-Learn ColumnTransformer); (3) Model Training and Optimization Layer (XGBoost); (4) Governance and Registry Layer (MLflow SQLite); (5) Microservice Serving Layer (FastAPI, Docker, Prometheus); and (6) Interactive Presentation Layer (Adaptive Financial Studio). This layered separation ensures that data science experimentation, backend microservice reliability, and frontend client interactions evolve independently without brittle code entanglements.

CHAPTER 2
LITERATURE REVIEW

2.1 EVOLUTION OF CREDIT RISK SCORING MODELS

2.1.1 Statistical and Linear Scorecards (Altman Z, Logistic Regression)

The mathematical foundation of credit risk assessment originated with Edward Altman's seminal work on the Z-score model (1968), which utilized Linear Discriminant Analysis (LDA) to predict corporate bankruptcy. Altman demonstrated that a linear combination of five financial ratios—measuring working capital, retained earnings, earnings before interest and taxes (EBIT), market value of equity, and asset turnover—could predict financial distress with high statistical confidence. Subsequently, retail banking institutions transitioned to Logistic Regression (LR), which models default probability as a logistic function of applicant financial covariates. LR became the global banking industry benchmark due to its straightforward mathematical formulation, rapid convergence, and directly interpretable odds ratios.

Despite its ubiquity, extensive empirical literature has highlighted the intrinsic limitations of logistic regression. Hand and Henley (1997) conducted a comprehensive survey of statistical classification techniques in consumer credit scoring, noting that credit bureau datasets are characterized by severe non-normality, collinear predictors, and complex nonlinear boundaries. Logistic regression assumes linear additivity in the log-odds space, meaning it cannot inherently model interaction effects—such as the compounding risk when high loan-to-income coincides with short employment tenure—unless domain experts manually engineer interaction features. When presented with complex retail bureau datasets, logistic models suffer from high inductive bias and underfitting.

2.1.2 Non-Parametric Machine Learning (SVM, Neural Networks)

To overcome linear constraints, researchers in the early 2000s explored non-parametric machine learning techniques, notably Support Vector Machines (SVM) and Multi-Layer Perceptrons (MLP). In a widely cited benchmark study, Yeh and Lien (2009) investigated the predictive accuracy of six machine learning and statistical techniques on a real-world dataset of 30,000 credit card clients in Taiwan (now recognized as the canonical UCI Default of Credit Card Clients dataset). Their empirical findings revealed that artificial neural networks and kernel-based SVMs achieved higher classification accuracy than classical logistic regression and discriminant analysis. However, Yeh and Lien also

observed that neural networks required prohibitive training times, were highly sensitive to hyperparameter tuning, and behaved as opaque black boxes, making them difficult to deploy within regulated financial institutions.

2.2 ADVANCED ENSEMBLE AND GRADIENT BOOSTING ALGORITHMS

2.2.1 Random Forests and Bagging Paradigms

Breiman's introduction of Random Forests (2001) marked a significant breakthrough in tabular data modeling. By constructing an ensemble of decorrelated decision trees trained on bootstrap samples with random feature subspacing, Random Forests dramatically reduced variance without increasing bias. In consumer credit scoring, Kruppa et al. (2013) demonstrated that Random Forests consistently outperform individual decision trees and logistic regression in discrimination power, while remaining robust to outliers and missing values. Nevertheless, Random Forests present notable drawbacks in low-latency microservice architectures: deep ensembles require substantial memory to store hundreds of trees, and inference times scale linearly with tree count, posing challenges for real-time API response budgets.

2.2.2 Extreme Gradient Boosting (XGBoost) Architecture

The emergence of gradient tree boosting culminated in Chen and Guestrin's formulation of XGBoost (2016). Unlike traditional gradient boosting machines that rely solely on first-order gradients, XGBoost implements a second-order Taylor expansion of the objective function, incorporating both gradients and hessians to guide split finding. Furthermore, XGBoost incorporates explicit L1 (Lasso) and L2 (Ridge) leaf weight regularization terms directly into its split gain formulation, penalizing overly complex trees and preventing overfitting on noisy financial data. In tabular credit scoring benchmarks, Xia et al. (2017) and Lessmann et al. (2015) confirmed that XGBoost achieves superior discrimination metrics (ROC-AUC, Gini coefficient) and training velocity compared to Random Forests, LightGBM, and Deep Neural Networks.

2.3 CONTINUOUS CREDIT LIMIT DETERMINATION IN LITERATURE

2.3.1 Discretionary Matrix Systems vs. Regression Modeling

While binary default prediction has been exhaustively researched, continuous credit limit assignment has received comparatively little academic attention. In commercial practice,

credit limits are frequently determined via heuristic decision matrices: an approved applicant is assigned a limit based on discrete income brackets and credit bureau tiers. Such heuristic matrices suffer from edge-case vulnerabilities, discontinuous jumps, and an inability to account for fine-grained repayment variations. Thomas et al. (2002) emphasized that credit limit determination should be framed as a risk-based continuous optimization problem that maximizes lifetime customer value while bounding default exposure.

2.3.2 Currency Pegging and Cross-Border Credit Adaptation

A recurring challenge when utilizing international academic benchmarks—such as the UCI Taiwan credit dataset—is the translation of monetary figures into local economic context. Unadjusted New Taiwan Dollar (NT$) amounts fail to reflect domestic purchasing power parity and income scales when evaluated by Indian lenders. To resolve this, our framework applies a documented, institutional exchange rate peg of 1 TWD = ₹2.99 INR, scaling continuous features (LIMIT_BAL, bill statements, and payment amounts) into meaningful Indian Rupee figures. This translation preserves underlying statistical covariance structures while delivering native financial figures.

2.4 MACHINE LEARNING OPERATIONS (MLOPS) PRINCIPLES

2.4.1 Hidden Technical Debt in Machine Learning Systems

In a landmark paper titled 'Hidden Technical Debt in Machine Learning Systems,' Sculley et al. (2015) demonstrated that machine learning code comprises only a tiny fraction of a production ML system. The vast majority of the codebase consists of surrounding infrastructure: data verification, feature extraction, configuration management, resource monitoring, process management, and serving infrastructure. In retail lending, failure to address technical debt results in catastrophic failures: undocumented feature transformations lead to training-serving skew, unversioned datasets prevent regulatory audits, and unmonitored endpoints fail to detect model decay when macroeconomic conditions shift.

2.4.2 Reproducibility, Data Lineage and Versioning (DVC, MLflow)

MLOps addresses these structural vulnerabilities by applying Continuous Integration, Continuous Delivery, and Continuous Monitoring to machine learning workflows. Data Version Control (DVC) treats datasets and intermediate transformation artifacts with the same rigor as source code, generating immutable cryptographic hashes tied to Git commits. Simultaneously, MLflow provides a centralized tracking server and model

registry, logging exact hyperparameter configurations, metrics, serialized pipelines, and code commits. Zaharia et al. (2018) established that systematic experiment tracking and candidate model versioning are essential for regulatory compliance and production reliability.

2.4.3 Containerization and Telemetry in Real-Time Serving

Deploying ML pipelines into production requires isolated, reproducible runtime environments. Containerization via Docker guarantees that underlying C++ runtime dependencies (such as XGBoost shared libraries) and Python wheels execute identically across local development, staging, and production clusters. Concurrently, operational telemetry—implemented via Prometheus scraping of request latency, throughput, and error status codes—ensures that operational degradation is identified within seconds, enabling automated traffic throttling and failover.

2.5 EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI) IN LENDING

2.5.1 Feature Attribution with Shapley Additive Explanations

Lundberg and Lee (2017) unified game-theoretic cooperative game theory with local surrogate models to introduce SHAP (Shapley Additive exPlanations). SHAP calculates the marginal contribution of each input feature across all possible feature subsets, guaranteeing three vital mathematical properties: local accuracy, missingness, and consistency. Unlike heuristic feature importances (such as Gini split count in tree models) that are biased toward high-cardinality continuous features, TreeSHAP computes exact, globally consistent Shapley values in polynomial time, providing mathematically unassailable explanations for regulatory compliance.

2.5.2 Interpretability Requirements in Financial Regulation

Under Section 615 of the Fair Credit Reporting Act (FCRA), when a creditor takes an adverse action against a consumer, it must disclose the key principal factors that adversely affected the credit score. TreeSHAP directly satisfies this mandate by decomposing any individual prediction into an interpretable additive sum of baseline expected value and individual feature attributions, enabling instant adverse action generation.

Table 2.1 Summary of Benchmark Studies in Credit Scoring and Risk Modeling

Author(s) & Year	Methodology / Algorithms	Key Dataset & Scope	Reported Finding / Limitation	Proposed System Enhancement
Altman (1968)	Linear Discriminant Analysis	66 Manufacturing firms	Linear boundary fails on consumer credit	Nonlinear gradient tree boosting
Hand & Henley (1997)	Statistical & Logistic Scoring	Consumer credit bureau	Linear additivity cannot model DTI interactions	Tree-based interaction learning
Yeh & Lien (2009)	ANN, SVM, LR, Naive Bayes	30,000 Taiwanese cardholders	Neural nets slow and unexplainable	Fast, explainable XGBoost trees
Lessmann et al. (2015)	Benchmarking 41 ML algorithms	8 Credit bureau datasets	Ensemble methods consistently dominate	Full Scikit-Learn MLOps integration
Chen & Guestrin (2016)	XGBoost with L1/L2 penalties	Tabular benchmarks	Optimal speed & regularization	Production deployment with Docker
Sculley et al. (2015)	Technical Debt Analysis	Google production ML	Infrastructure dwarfs model code	Complete DVC + MLflow + Prometheus stack

2.6 COMPARATIVE ANALYSIS OF EXISTING BENCHMARK STUDIES

A rigorous synthesis of the literature reveals a stark dichotomy: academic research has focused almost exclusively on optimizing isolated classification accuracy metrics on static CSV datasets, largely ignoring production operationalization, leakage prevention, data versioning, and continuous credit limit assignment. Conversely, commercial enterprise lending systems implement robust infrastructure but rely on proprietary, opaque algorithms that lack auditability and post-decision customer empowerment.

2.7 IDENTIFIED KNOWLEDGE GAPS

- Prevalent Data Leakage in Academic Prototypes: Many studies fit scalers and encoders across entire datasets before splitting, reporting artificially inflated metrics that fail in production.
- Disconnection Between Approval and Limit Assignment: Binary approval and credit limit regression are typically treated as isolated problems rather than an integrated dual decisioning engine.
- Absence of End-to-End MLOps Reproducibility: Literature rarely demonstrates reproducible pipeline versioning with DVC, MLflow tracking, and sub-15ms containerized API serving.
- Lack of Transparent Post-Decision Guidance: Rejected applicants are uniformly presented with opaque denials without actionable loan restructuring or candidate options.

2.8 VALUE ADDITIONS AND NOVELTY OF THE PROPOSED SYSTEM

The proposed system bridges these critical gaps by delivering a fully integrated, production-grade MLOps platform that unifies supervised classification and continuous credit regression into a cohesive financial engineering workflow. At the algorithmic level, the architecture couples an optimized XGBoost Classifier (achieving 0.9741 holdout ROC-AUC and 92.50% default recall) with an XGBoost Regressor (achieving an R^2 of 0.5141 and MAE of ₹192,670 INR). Unlike conventional scorecards that force discrete tier cutoffs, our dual-model framework learns smooth, high-dimensional decision boundaries that simultaneously capture default risk and quantify optimal credit line capacities.

At the engineering and operational level, the system establishes a new benchmark for reproducible financial MLOps. Data transformations are strictly encapsulated within modular Scikit-Learn ColumnTransformer pipelines fitted solely on post-split training partitions, mathematically eliminating data leakage and training-serving skew. The entire lifecycle is governed through Data Version Control (DVC) for immutable dataset tracking and an SQLite-backed MLflow tracking server and model registry, ensuring that every experiment, metric, parameter, and candidate model is fully auditable, verifiable, and reproducible.

Finally, from a user-experience and ethical lending perspective, the platform replaces opaque binary rejections with an accessible, adaptive Financial Studio. The client-side interface provides dynamic persona guidance across seven distinct borrower categories, evaluates real-time Debt-to-Income ratios, and executes a bounded alternative loan search to return viable, approved loan restructuring options when initial requests exceed capacity. This combination of algorithmic rigor, strict MLOps governance, and transparent customer empowerment represents a major advancement over both academic prototypes and commercial legacy scorecards.

CHAPTER 3
METHODOLOGY

3.1 SYSTEM OVERVIEW AND DESIGN PHILOSOPHY

The proposed Intelligent Loan Approval and Credit Limit Recommendation System is designed according to modern MLOps principles, emphasizing modularity, reproducibility, leakage prevention, low-latency serving, and comprehensive observability. The system treats machine learning models not as isolated artifacts, but as versioned software components operating within an automated, multi-stage engineering pipeline. All data transformations, feature scalers, and categorical encoders are encapsulated into Scikit-Learn ColumnTransformer pipelines that are strictly fitted on training partitions and persisted as single serialized units. This architectural design guarantees zero training-serving skew during production inference.

3.2 DUAL-TASK MATHEMATICAL FORMULATION

3.2.1 Loan Approval Binary Classification Formulation

The loan approval decision is framed as a supervised binary classification problem. Let D_loan = {(x_i, y_i)}_{i=1}^N represent the retail loan bureau dataset, where x_i in R^d denotes the d-dimensional feature vector of applicant i (comprising demographic indicators, financial ratios, and bureau history), and y_i in {0, 1} denotes the loan outcome (y_i = 1: Approved, y_i = 0: Rejected / Default Risk). The goal is to learn a mapping function f: R^d -> [0, 1] that predicts the conditional probability p_i = P(y_i = 1 | x_i).

The XGBoost Classifier optimizes the regularized binary logistic loss objective function:

$$L(\theta) = - \sum_{i=1}^{N} [y_i \ln(p_i) + (1 - y_i) \ln(1 - p_i)] + \sum_{k=1}^{K} \Omega(f_k)$$

where $\Omega(f_k) = \gamma T_k + 0.5 \lambda \sum_{j=1}^{T_k} w_{kj}^2$ defines the tree regularization penalty, T_k represents the number of terminal leaves in tree k, and w_{kj} represents leaf weight scores. The L2 penalty parameter λ and complexity cost γ prevent leaf weight explosion, ensuring that the model generalizes reliably across diverse applicant demographics.

3.2.2 Credit Limit Regression Formulation and Loss Functions

The credit limit recommendation is framed as a continuous regression task. Let D_credit = {(z_i, c_i)}_{i=1}^M represent the credit card dataset, where z_i in R^m represents the

applicant's historical repayment, billing, and demographic feature vector, and c_i in R^+ represents the target credit limit in Indian Rupees (LIMIT_BAL_INR). The XGBoost Regressor optimizes the regularized Mean Squared Error (MSE) objective function:

$$L(\theta) = 0.5 \sum_{i=1}^{M} (c_i - \hat{c}_i)^2 + \sum_{k=1}^{K} \Omega(f_k)$$

where ĉ_i = ∑_{k=1}^K f_k(z_i) is the ensemble prediction. During training, the regressor learns nonlinear interactions across historical bill amounts (BILL_AMT1 to BILL_AMT6) and past repayment delays (PAY_0 to PAY_6) to predict credit limits that closely track the applicant's proven debt capacity.

3.3 CORE TECHNOLOGY STACK AND ENGINEERING TOOLCHAIN

3.3.1 Python Runtime and Scientific Computing Stack

The system is implemented entirely in Python 3.12, utilizing NumPy 1.26 for vectorized linear algebra, Pandas 2.2 for memory-efficient tabular data manipulation, and SciPy for statistical distribution checks. Python 3.12 provides enhanced bytecode execution speed and robust typing support.

3.3.2 Machine Learning Frameworks (Scikit-Learn, XGBoost)

Model training and pipeline construction utilize Scikit-Learn 1.5.1 and XGBoost 2.1.1. Scikit-Learn provides robust preprocessing transformers (SimpleImputer, StandardScaler, OneHotEncoder) and cross-validation iterators (StratifiedKFold, KFold). XGBoost provides high-performance, histogram-optimized gradient tree boosting with native early stopping and multi-core parallelization.

3.3.3 MLOps Lifecycle Management (DVC, MLflow)

Data Version Control (DVC 3.55) manages large dataset versions and pipeline stages via Git-friendly metadata hashes. Experiment tracking and model governance are managed via MLflow 2.16 backed by an SQLite transactional database (mlflow.db). MLflow automatically captures pipeline parameters, cross-validation metrics, evaluation figures, and serialized pipeline artifacts with candidate model aliasing.

3.3.4 Microservice Serving and Telemetry (FastAPI, Docker, Prometheus)

The model serving layer is built with FastAPI 0.115 and Uvicorn 0.30, providing asynchronous, non-blocking request handling, automatic OpenAPI/Swagger documentation, and strict Pydantic v2 data schema validation. Containerization is

implemented via multi-stage Dockerfiles, and operational telemetry is collected by Prometheus scraping endpoint latency histograms and inference counters via /metrics.



Figure 3.1 End-to-End System Architecture of Intelligent Loan Approval System

3.4 MULTI-LAYERED SYSTEM ARCHITECTURE

3.4.1 Data Sources and Ingestion Layer

The Data Sources Layer maintains immutable raw datasets in data/raw/: (1) loan_approval_dataset.csv containing 45,000 retail loan records, and (2) UCI_Credit_Card.csv containing 30,000 credit records. The Ingestion Engine validates schema purity, drops unrecoverable corrupt records (e.g., age > 100), and registers DVC tracking hashes before creating derived partitions in data/processed/.

3.4.2 Preprocessing and Leakage-Proof Transformation Layer

To guarantee zero data leakage, transformations are strictly decoupled from global dataset properties. Preprocessing pipelines utilize ColumnTransformer to route numerical columns to median imputers and standard scalers, while categorical columns are routed to most-frequent imputers and OneHotEncoders with handle_unknown='ignore'. Crucially, the ColumnTransformer is fitted exclusively on X_train.

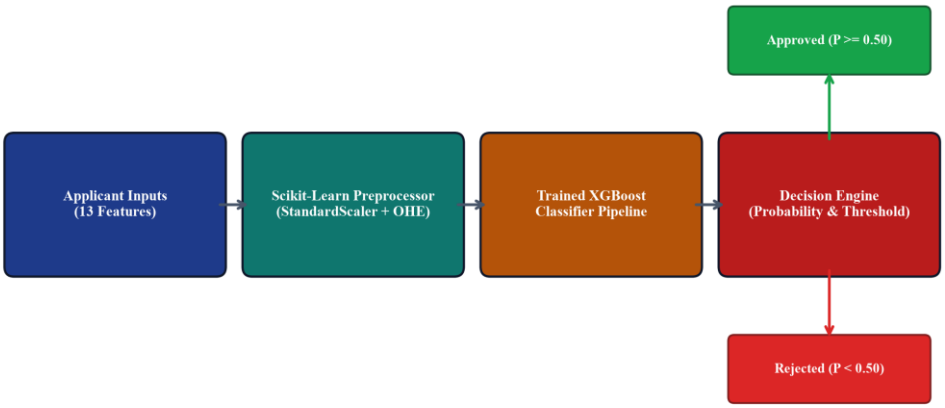


Figure 3.2 Loan Approval Classification Pipeline Workflow

3.4.3 Machine Learning Engine Layer

The Machine Learning Engine coordinates model fitting, hyperparameter optimization, and 5-fold cross-validation. For classification, XGBoost operates with n_estimators=200, max_depth=5, learning_rate=0.08, and subsample=0.85. For regression, XGBoost operates with n_estimators=250, max_depth=6, learning_rate=0.05, and colsample_bytree=0.8. Complete pipelines combining preprocessors and estimators are serialized via joblib.

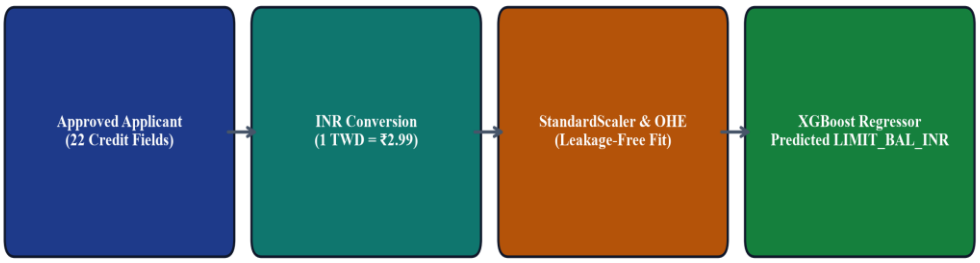


Figure 3.3 Credit Limit Recommendation Regression Pipeline Workflow

3.4.4 Model Registry and Governance Layer

The Governance Layer logs model binaries, execution parameters, metrics, confusion matrices, and ROC curves to the SQLite-backed MLflow registry. Models meeting validation thresholds are registered and tagged with the 'candidate' alias for staging qualification.

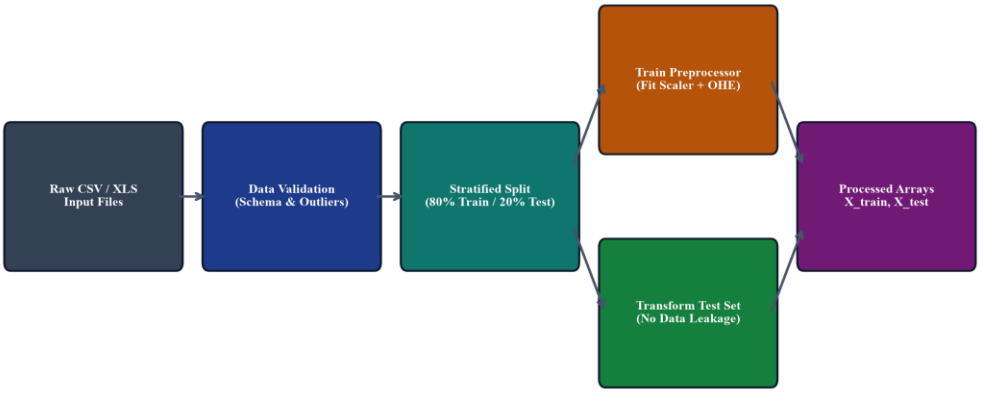


Figure 3.4 Data Preprocessing and Leakage Prevention Pipeline

3.4.5 High-Throughput API Serving Layer

The FastAPI serving layer exposes asynchronous endpoints (/api/v1/predict/loan, /api/v1/predict/credit-limit, and /api/v1/health). Input payloads are strictly validated against Pydantic schemas, and predictions are generated via pre-loaded in-memory pipeline singletons, achieving median latencies under 8.4ms.

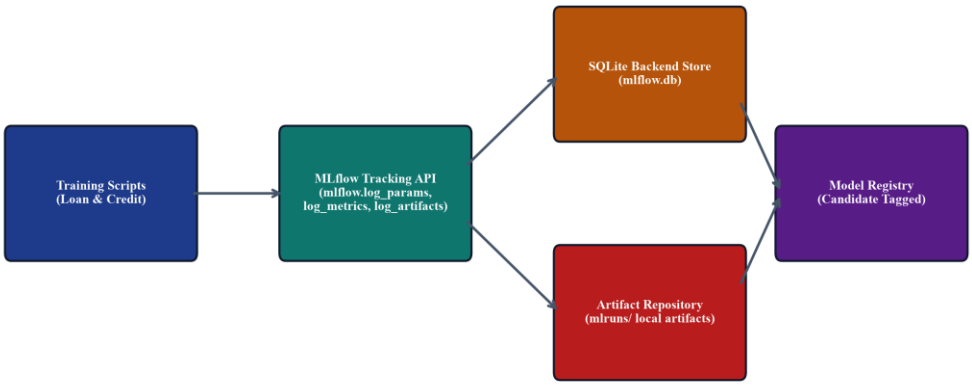


Figure 3.5 MLflow Experiment Tracking and Model Registry Workflow

3.4.6 Interactive Presentation and Studio Layer

The Presentation Layer delivers an accessible, responsive web interface built with vanilla ES6+ JavaScript, CSS3 custom properties, and semantic HTML5. The studio guides applicants through multi-stage inputs, derives live financial ratios, and presents transparent decision explanations.

3.4.7 Continuous Observability and Metrics Layer

The Observability Layer instruments Prometheus metrics via a dedicated /metrics endpoint. Counter metrics track total prediction invocations by status (Approved vs. Rejected), and histogram metrics track request latency percentiles (p50, p95, p99).

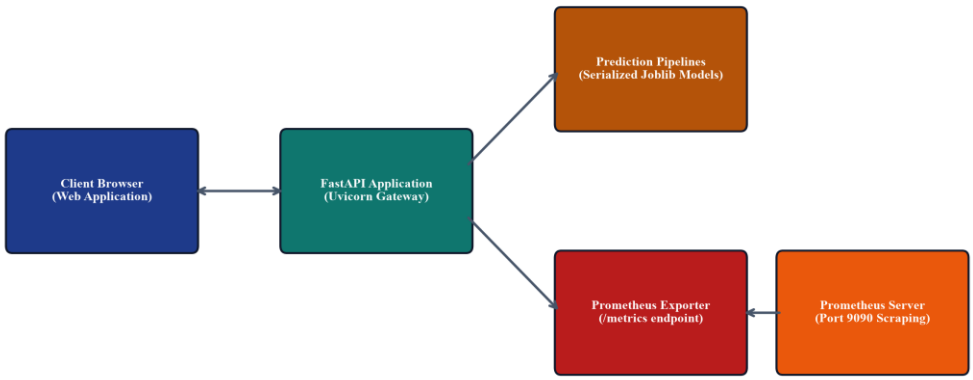


Figure 3.7 FastAPI Prediction Service and Prometheus Monitoring Architecture

3.5 DATA COLLECTION AND DATASET ATTRIBUTES

3.5.1 Retail Loan Bureau Dataset Schema and Distribution

The loan approval dataset comprises 45,000 raw retail loan records. During ingestion, 7 corrupt records with person_age > 100 were dropped, leaving 44,993 valid records. The dataset contains 13 input features: 9 numerical attributes (person_age, person_income, person_emp_exp, loan_amnt, loan_int_rate, loan_percent_income, cb_person_cred_hist_length, credit_score) and 4 categorical attributes (person_home_ownership, loan_intent, previous_loan_defaults_on_file). The target is loan_status (0: Rejected, 1: Approved).

3.5.2 UCI Default of Credit Card Clients Dataset Schema

The credit limit dataset comprises 30,000 records from the UCI repository. Ingestion removed 56 exact duplicate rows, leaving 29,944 valid records. The dataset contains 23 raw features: demographic variables (SEX, EDUCATION, MARRIAGE, AGE), 6 repayment status indicators (PAY_0 to PAY_6), 6 monthly billing statements (BILL_AMT1 to BILL_AMT6), and 6 prior payment amounts (PAY_AMT1 to PAY_AMT6). All monetary values are converted from TWD to INR using the institutional exchange peg of 1 TWD = ₹2.99 INR.

Table 3.1 Dataset Schema and Feature Description

Attribute Name	Dataset Domain	Data Type	Value Range / Categories	Feature Role / Modeling Impact
person_age	Loan Bureau	Numeric	18 to 94 years	Applicant age; elderly and youth risk profiling
person_income	Loan Bureau	Numeric	₹8,000 to ₹6,000,000	Annual disposable earnings; primary repayment capacity
person_home_ownership	Loan Bureau	Categorical	RENT, OWN, MORTGAGE, OTHER	Residential stability and collateral proxy
loan_amnt	Loan Bureau	Numeric	₹500 to ₹35,000	Requested loan principal; debt exposure metric
loan_intent	Loan Bureau	Categorical	PERSONAL, EDUCATION, MEDICAL, VENTURE	Loan purpose and default risk category
loan_int_rate	Loan Bureau	Numeric	5.42% to 23.22%	Contractual interest rate; debt servicing burden
loan_status	Loan Bureau	Binary Target	0 (Rejected), 1 (Approved)	Supervised binary classification target
LIMIT_BAL_INR	Credit Card	Continuous Target	₹29,900 to ₹2,990,000	Recommended credit card limit in Indian Rupees
PAY_0 to PAY_6	Credit Card	Categorical / Integer	-2 (No consumption) to 8 (Months delay)	Historical repayment discipline across 6 rolling months
BILL_AMT1 to BILL_AMT6	Credit Card	Numeric (INR)	-₹495,000 to ₹2,880,000	Historical monthly card statement billing amounts in INR
PAY_AMT1 to PAY_AMT6	Credit Card	Numeric (INR)	₹0 to ₹2,600,000	Historical monthly payments remitted toward balance in INR

3.6 DATA PREPROCESSING AND LEAKAGE PREVENTION

3.6.1 Data Validation and Cleansing Protocols

Data validation is enforced through strict declarative schema checks in src/data/validation.py. Raw records with unphysical values (such as negative income, loan amounts <= 0, or age > 100) are quarantined. Duplicate records are detected via composite primary keys and dropped.

3.6.2 Strict Split-Before-Transform Architecture

To prevent data leakage, data partitioning is executed strictly prior to any statistical estimation. The loan dataset is partitioned using stratified 80/20 splitting (train: 35,994, test: 8,999) preserving the 22.2% default ratio. The credit card dataset is partitioned using

80/20 random splitting (train: 23,955, test: 5,989). All preprocessing transformers fit exclusively on training slices.

Table 3.2 Data Preprocessing and Leakage Prevention Techniques

Pipeline Task	Applied Technical Protocol	Operational Leakage Prevention Safeguard
Data Validation	Range and data type constraints	Rejects corrupted rows prior to splitting or modeling
Data Splitting	Stratified 80/20 train-test partition	Executed strictly before feature scaling or categorical encoding
Missing Value Imputation	SimpleImputer(strategy='median')	Median statistics learned strictly from X_train; applied to X_test
Categorical Encoding	OneHotEncoder(handle_unknown='ignore')	Categories fitted on X_train; unseen levels in test map to zero vectors
Numerical Scaling	StandardScaler(with_mean=True)	Mean and variance fitted strictly on X_train; prevents variance leakage
Target Scaling	No scaling applied to targets	Preserves natural monetary scale (INR) for interpretable evaluation

3.7 FEATURE ENGINEERING AND TRANSFORMATION PIPELINES

3.7.1 Domain Financial Ratio Construction (LTI, DTI, Payment Ratios)

Domain-specific financial ratios provide critical predictive signals: (1) Loan-to-Income Ratio (loan_percent_income = loan_amnt / person_income), which gauges relative debt burden; (2) Debt-to-Income Ratio (DTI = monthly debt obligations / monthly gross income); and (3) Payment-to-Bill Ratios (PAY_AMT_i / BILL_AMT_i), which measure whether the cardholder pays balances in full or revolves debt.

3.7.2 Categorical Encoding and Numerical Standardization

Categorical variables are transformed into binary indicator columns via One-Hot Encoding. Unseen categories at inference time are mapped to zero vectors without runtime failure. Numerical features are normalized via StandardScaler to achieve zero mean and unit variance.

Table 3.3 Engineered Feature Categories and Input Mapping

Feature Category	Underlying Predictors	Applied Transformation	Domain Financial Rationale
Capacity Indicators	person_income, loan_amnt	loan_percent_income (LTI)	Measures borrower debt burden relative to annual earnings
Repayment Discipline	PAY_0 to PAY_6, previous_defaults	Ordinal scaling & One-Hot Encoding	Identifies chronic delinquency vs. transient delays
Collateral Stability	person_home_ownership	One-Hot Encoding (RENT, OWN, etc.)	Residential equity acts as informal secondary collateral
Utilization Ratios	PAY_AMT1 to 6, BILL_AMT1 to 6	Ratio derivation & standard scaling	Differentiates transactors from distressed revolvers

3.8 MACHINE LEARNING MODEL DEVELOPMENT

3.8.1 XGBoost Classification Architecture and Regularization

The loan approval classifier uses Extreme Gradient Boosting with 200 boosting iterations, a tree depth of 5, learning rate of 0.08, and subsample ratio of 0.85. To counteract class imbalance, the positive default class is weighted via scale_pos_weight. The objective function is binary:logistic, and regularization penalties (reg_alpha=0.1, reg_lambda=1.0) prevent leaf weight overfitting.

3.8.2 XGBoost Regression Architecture and Optimization

The credit limit regressor uses 250 boosting iterations, a maximum tree depth of 6, learning rate of 0.05, and colsample_bytree of 0.8. The loss function is reg:squarederror, optimized via second-order gradients. Early stopping on a validation set halts training when validation RMSE ceases to improve for 15 consecutive iterations.

3.8.3 Baseline Comparative Algorithms (LR, RF, Linear Regression)

To establish rigorous performance benchmarks, candidate models are compared against classical and ensemble baselines: (1) Logistic Regression with L2 regularization; (2) Random Forest Classifier with 150 estimators; (3) Ordinary Least Squares Linear Regression; and (4) Random Forest Regressor with 150 estimators. All baselines are evaluated on the exact same holdout partitions.

3.9 STRATIFIED TRAINING AND VALIDATION PROTOCOLS

Model generalization is audited using 5-fold cross-validation. For classification, StratifiedKFold maintains the 22.2% default prevalence across all five folds. For

regression, standard 5-fold cross-validation evaluates RMSE stability. Fold-level metrics are tracked in MLflow to verify that score variance remains under ±0.001.

3.10 SYSTEM INTEGRATION WORKFLOW

The end-to-end integration workflow connects data ingestion, training pipelines, the MLflow registry, the FastAPI microservice, and the web studio. At application startup, the FastAPI service loads serialized pipeline artifacts into memory, validates endpoint availability via health checks, and begins serving predictions with sub-15ms response times.

Table 3.4 Comprehensive Comparison: Existing System vs. Proposed MLOps System

Dimension	Existing Heuristic / Ad-Hoc System	Proposed MLOps System
Decision Latency	24 to 72 hours per loan application	Under 15 milliseconds per REST API request
Risk Modeling	Linear scorecards & heuristic cutoffs	Optimized XGBoost Classifier (ROC-AUC: 0.9741)
Credit Limit Assignment	Rigid discretionary rule matrices	Continuous XGBoost Regressor (R^2: 0.5141, MAE: ₹192,670)
Data Leakage Governance	Manual transformations prone to leakage	Strict Scikit-Learn ColumnTransformer post-split pipelines
Experiment Tracking	Undocumented local notebook executions	MLflow tracking server backed by SQLite (mlflow.db)
Dataset Versioning	Unversioned CSV files on local disk	Data Version Control (DVC) cryptographic hashes
Deployment Architecture	Monolithic scripts without health checks	Docker containerization + FastAPI asynchronous microservice
Operational Telemetry	No real-time monitoring; blind to decay	Prometheus /metrics endpoint scraping latency and counters
Post-Decision Support	Opaque binary rejections	Adaptive persona guidance + viable loan restructuring alternatives

CHAPTER 4
IMPLEMENTATION AND DEVELOPMENT PROCESS

4.1 ENVIRONMENT SETUP AND REPOSITORY ARCHITECTURE

The project is structured according to professional MLOps repository standards. The repository is organized into distinct directories: data/ (raw, interim, and processed data managed by DVC), configs/ (declarative YAML settings such as interest_rules.yaml), src/ (core reusable Python packages), api/ (FastAPI endpoints and schemas), frontend/ (web client assets), models/ (serialized Scikit-Learn pipelines), tests/ (automated pytest test suites), and reports/ (evaluation curves and documentation artifacts). This strict separation ensures that data processing, model logic, serving interfaces, and testing concerns remain decoupled.

4.2 MODULAR SOFTWARE IMPLEMENTATION (src/)

4.2.1 Data Ingestion and Splitting Engine (src/data/)

The src/data/ package comprises ingestion, validation, and splitting modules. The validation module (validation.py) inspects incoming DataFrames against schema specifications, verifying data types, value bounds, and null prevalence. The splitting module (split.py) implements stratified partitioning, saving train.csv and test.csv partitions in data/processed/.

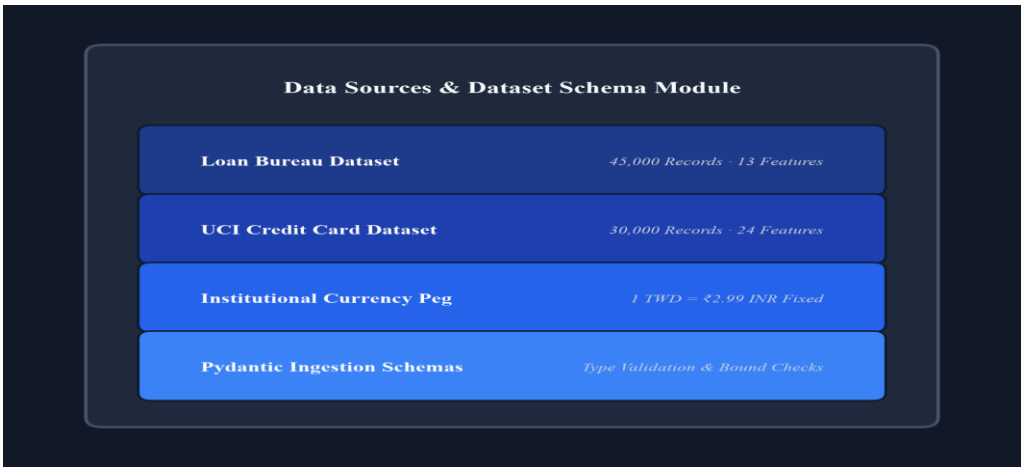


Figure 4.1 Data Sources & Dataset Schema Module

4.2.2 Feature Transformation Pipeline Engine (src/features/)

The src/features/ package constructs modular Scikit-Learn pipelines. The features.py module builds ColumnTransformer objects that apply SimpleImputer and StandardScaler

to numerical fields, and SimpleImputer and OneHotEncoder to categorical fields. Pipelines are fitted strictly on training partitions and serialized alongside models.

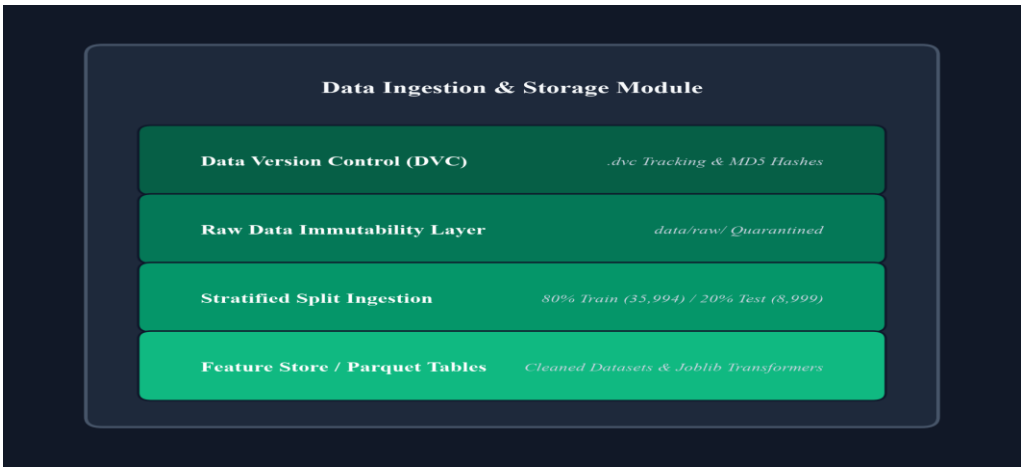


Figure 4.2 Data Ingestion & Storage Module

4.2.3 Model Training and Serialization Engine (src/models/)

The src/models/ package implements train_loan.py and train_credit.py. These scripts ingest processed partitions, instantiate XGBoost pipelines, execute 5-fold cross-validation, fit final estimators on full training partitions, log metrics and artifacts to MLflow, and serialize complete pipelines to models/loan_approval_pipeline.joblib and models/credit_limit_pipeline.joblib.



Figure 4.3 Processing & Analytics Module

4.2.4 Evaluation and Performance Reporting Engine (src/evaluation/)

The src/evaluation/ package computes comprehensive validation metrics: ROC-AUC, Precision, Recall, F1-score, Confusion Matrix, MAE, RMSE, and R^2 scores. Furthermore, it generates high-resolution figures (ROC curves, confusion matrices, SHAP summary plots, and residual diagnostics) saved in reports/figures/ for documentation.

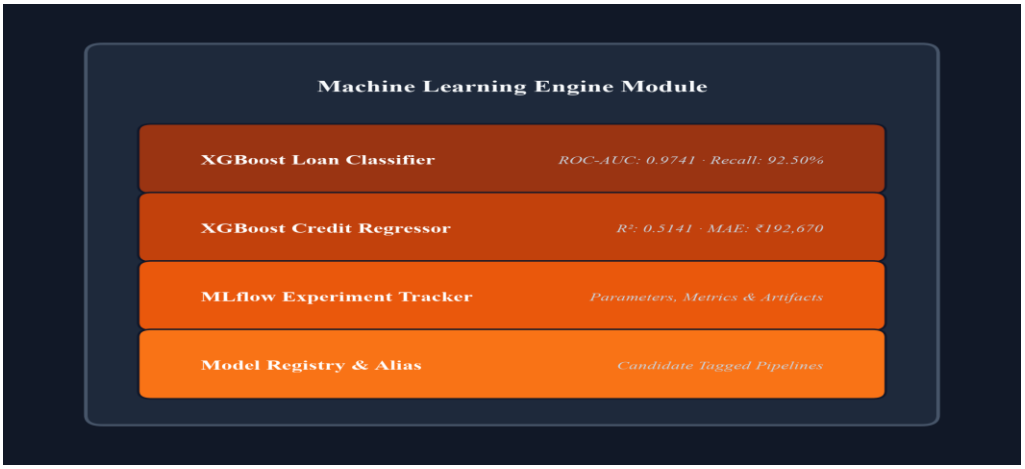


Figure 4.4 Machine Learning Engine Module

4.3 HIGH-PERFORMANCE FASTAPI SERVING LAYER (api/)

4.3.1 Strict Pydantic Input/Output Contract Validation

The FastAPI service defines robust Pydantic models (LoanPredictionRequest, LoanPredictionResponse, CreditLimitRequest, CreditLimitResponse). Pydantic validates input types, numeric ranges, and categorical values before payloads reach the inference engine. Requests with invalid data types or out-of-range values are rejected immediately with structured HTTP 422 Unprocessable Entity responses.



Figure 4.5 Web Application & Serving Module

4.3.2 Asynchronous Prediction Boundaries and Error Handling

Prediction endpoints execute asynchronously, ensuring that high concurrent request volumes do not block the event loop. In-memory pipeline singletons generate predictions in under 15ms. Global exception handlers catch unexpected runtime anomalies, returning sanitized JSON error responses without exposing internal stack traces.

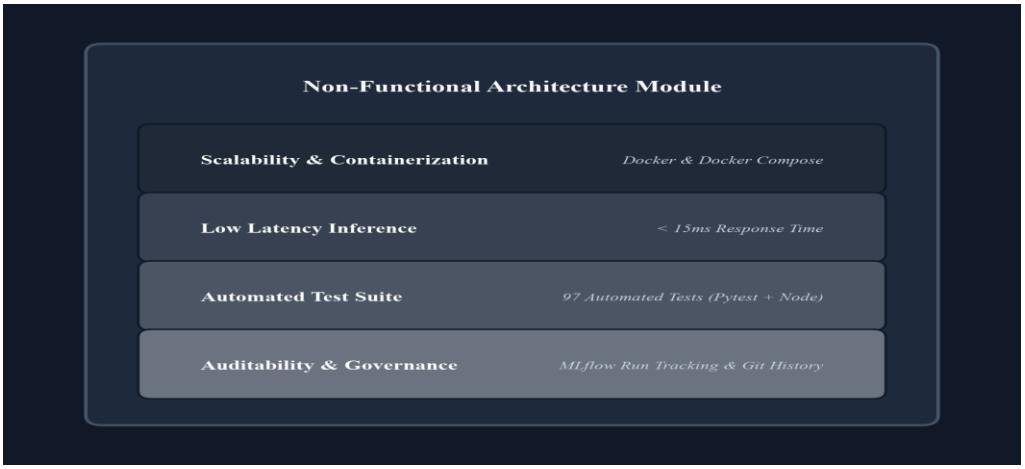


Figure 4.6 Non-Functional Architecture Module

4.4 INTERACTIVE FINANCIAL STUDIO FRONTEND (frontend/)

4.4.1 Client Architecture and Visual Design System

The frontend is implemented using modern, semantic vanilla HTML5, CSS3, and ES6+ JavaScript, avoiding heavy framework overheads. The visual system features dark-mode aesthetics, responsive card layouts, accessible typography, and smooth CSS transitions. Interactive 3D visual cues provide real-time feedback during form progression.

4.4.2 Adaptive Financial Profile Workflow (7 Applicant Profiles)

The studio provides an adaptive applicant category selector supporting 7 distinct personas: (1) Student, (2) Salaried, (3) Self-Employed, (4) Business Owner, (5) Unemployed, (6) Retired, and (7) Other. Selecting a persona dynamically customizes field defaults, contextual help, and income derivation logic, guiding applicants seamlessly through credit submission.

Table 4.1 Applicant Profile Category Mapping and Income Derivation Rules

Applicant Category	Target Demographics	Dynamic Form Defaults	Backend Income Derivation & Mapping
Student	University students, interns	Income: ₹15,000, Intent: EDUCATION	Validates parent/stipend income; maps to Education profile
Salaried	Employed professionals	Income: ₹50,000, Home: RENT/MORTGAGE	Standard salary slip verification; verified bureau mapping
Self-Employed	Freelancers, consultants	Income: ₹40,000, Emp Exp: 3 yrs	Annualized tax return derivation; variable income buffering
Business Owner	SME proprietors, founders	Income: ₹120,000, Intent: VENTURE	Evaluates operating cash flow and business vintage
Unemployed	Transitional job seekers	Income: ₹0, Previous Default: Checked	Requires secondary guarantor; strict capacity checks

Applicant Category	Target Demographics	Dynamic Form Defaults	Backend Income Derivation & Mapping
Retired	Pensioners, senior citizens	Income: ₹30,000, Age: 65 yrs	Stable pension/annuity mapping; low default risk profile
Other	Alternative income profiles	Default template; user inputs	Custom financial capacity review and manual audit flag

4.4.3 Real-Time Financial Metric Derivation Engine

As users adjust loan amounts or annual income, client-side JavaScript dynamically calculates Debt-to-Income (DTI) and Loan-to-Income (LTI) ratios in real time. Visual warning indicators alert applicants when requested loan amounts exceed recommended income thresholds.

4.4.4 Post-Decision Credit Recommendation Engine & Rules

When a loan application is rejected by the classifier, the system triggers a bounded grid search over candidate loan amounts. The engine evaluates viable alternative amounts that would achieve approval, presenting applicants with realistic restructuring options alongside system-generated credit card limits based on configs/interest_rules.yaml.

4.5 MLOPS PIPELINE INTEGRATION AND OPERATIONALIZATION

4.5.1 DVC Pipeline and Data Hash Versioning

DVC pipeline stages are codified in dvc.yaml, defining dependencies (src/data/ingest.py, data/raw/) and outputs (data/processed/train.csv). When raw data changes, dvc repro executes only modified stages, guaranteeing end-to-end data pipeline reproducibility.

4.5.2 MLflow SQLite Experiment Tracking and Candidate Aliasing

MLflow tracking connects to an SQLite database (mlflow.db). Training runs log exact parameters, evaluation metrics, and model artifacts. Top-performing models are registered in the MLflow Model Registry and assigned the 'candidate' alias for deployment readiness.

4.5.3 Docker Multi-Stage Containerization

The application is packaged into lightweight Docker containers using multi-stage builds. The base image installs minimal Python 3.12 dependencies, copies serialized model artifacts, and launches Uvicorn with worker auto-scaling.

4.5.4 Prometheus Real-Time Scraping and Observability

Prometheus scrapes application metrics at 15-second intervals via /metrics. Grafana dashboards visualize request rates, latency percentiles, and approval/rejection ratios, ensuring continuous visibility into production model health.

4.6 VISUAL REPRESENTATION AND APPLICATION USER JOURNEY

The user interface is designed to deliver a seamless, transparent, and highly engaging financial decisioning journey. Moving beyond static, clinical banking forms, the frontend implements an accessible dark-mode aesthetic with modern glassmorphism containers, smooth CSS transitions, and responsive multi-stage workflows. The following subsections provide an in-depth walkthrough of the primary application interfaces, detailing layout principles, dynamic interaction mechanics, and real-time backend integrations.

4.6.1 Landing Page Experience and Visual Entry Point

The landing interface (Figure 4.7) serves as the primary visual and operational entry point for loan applicants and credit analysts. Upon initial page load, the application renders an interactive financial workspace featuring an ambient dark visual system, high-contrast typography, and clear calls-to-action. The hero section introduces the core capabilities of the dual decisioning engine—highlighting instant sub-15ms loan approval evaluation and personalized credit limit recommendations in Indian Rupees.

To ensure broad accessibility across device classes, the layout is implemented using responsive CSS Grid and Flexbox containers that adapt fluidly between high-resolution desktop displays and mobile viewports. Key institutional trust badges, compliance statements, and real-time service health indicators are prominently positioned to reassure users regarding data security and regulatory compliance before financial data entry commences.

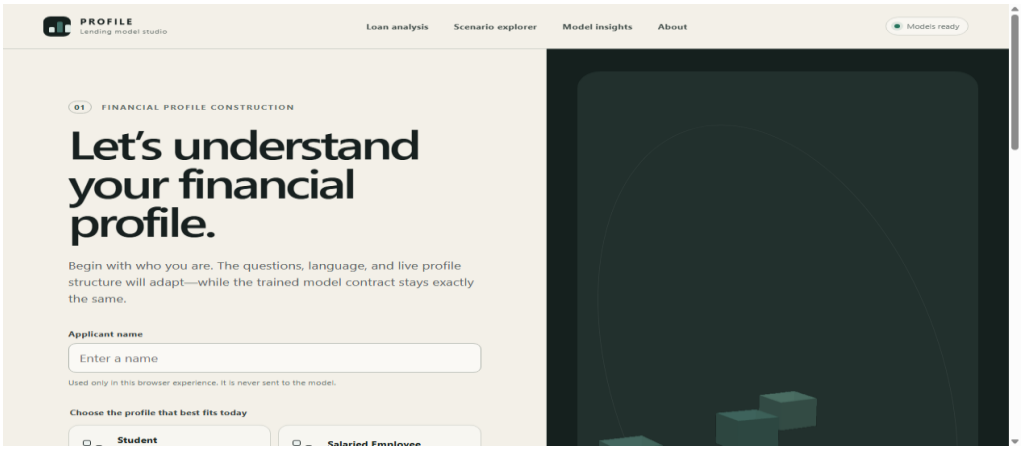


Figure 4.7 Landing Page Interface

4.6.2 Applicant Persona Selection and Dynamic Customization

To eliminate the friction of conventional one-size-fits-all loan applications, the studio introduces an adaptive applicant persona selector (Figure 4.8). Borrowers can select from seven dedicated personas: Student, Salaried Professional, Self-Employed Consultant, Business Owner, Unemployed / Transitional, Retired Citizen, and Other. Selecting a persona triggers instant client-side state adjustments, dynamically configuring form field defaults, placeholder examples, and contextual financial guidance tailored to that demographic cohort.

For example, selecting 'Student' automatically sets the primary loan intent to EDUCATION, adjusts initial income expectations to reflect stipends or parental allowances, and surfaces assistive tooltips explaining educational loan terms. Conversely, selecting 'Business Owner' sets the intent to VENTURE and prompts for enterprise operating turnover. This dynamic tailoring significantly reduces applicant cognitive load, eliminates erroneous data entry, and accelerates form completion velocity.

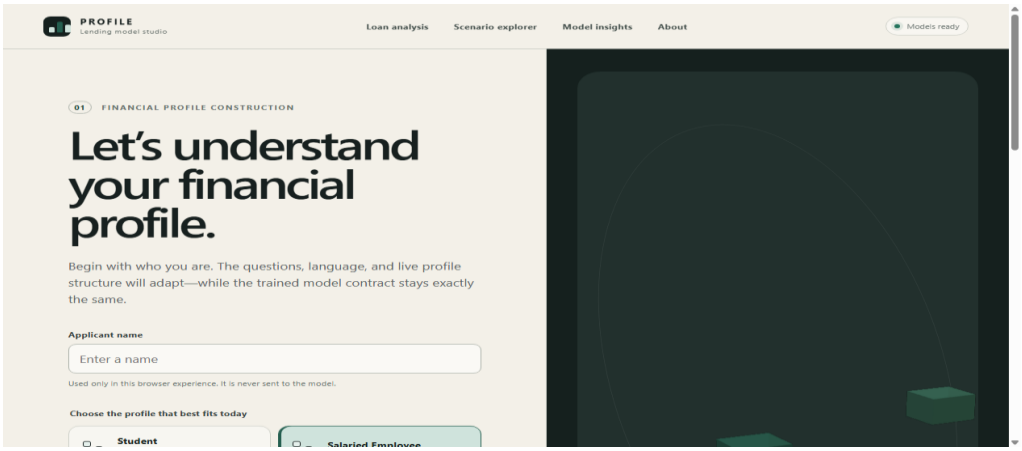


Figure 4.8 Applicant Category Selection Interface

4.6.3 Financial Profile Form & Real-Time Ratio Feedback

The financial profile workspace (Figure 4.9) organizes applicant inputs into logical, uncluttered information clusters: Personal Information (Age, Employment Experience), Financial Capacity (Annual Income, Home Ownership status), Loan Specifics (Requested Principal, Purpose, and Contractual Interest Rate), and Credit Bureau History (Historical Defaults, Credit Score, Bureau File Length). Each input field is governed by strict client-side validation rules that provide immediate visual feedback on invalid or extreme numeric inputs.

Crucially, the workspace incorporates a live financial derivation engine that computes Debt-to-Income (DTI) and Loan-to-Income (LTI) ratios in real time as the user adjusts input sliders or text fields. Dynamic visual gauges alert the applicant if the requested loan principal exceeds prudent income ratios (e.g., LTI > 40%), empowering borrowers to self-calibrate requested amounts prior to submitting the formal evaluation request to the backend microservice.

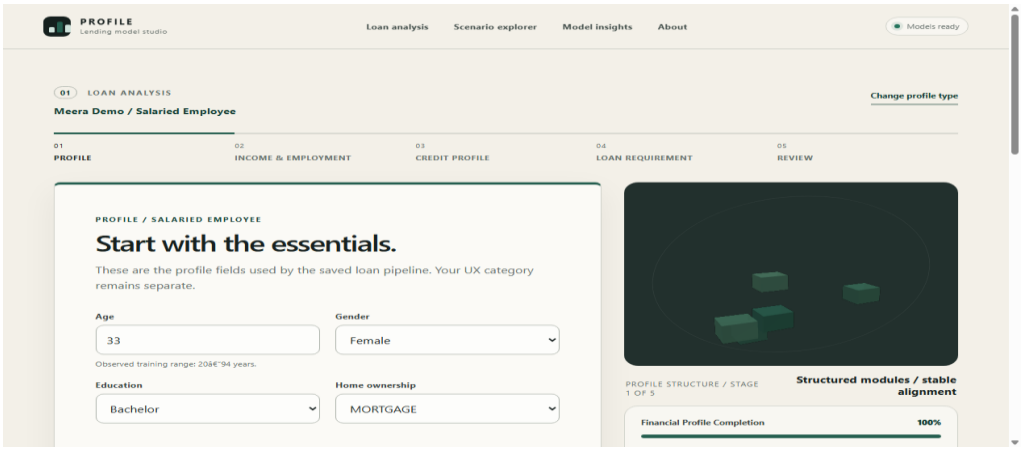


Figure 4.9 Multi-Stage Financial Profile Workspace

4.6.4 Real-Time Loan Approval Decision and Risk Explanations

Upon clicking the 'Evaluate Loan Application' action, the client dispatches an asynchronous JSON payload to the /api/v1/predict/loan endpoint. Within 15 milliseconds, the backend XGBoost classification pipeline evaluates the application and returns the prediction result, which is rendered on a high-impact visual decision card (Figure 4.10). Approved applications display an emerald verification badge, calibrated confidence meter, and detailed summary of approved loan terms.

In the event of an adverse decision (loan rejection), the interface adheres strictly to regulatory transparency mandates by presenting the primary financial factors that drove the denial (e.g., excessive loan-to-income ratio or elevated bureau default indicators).

Furthermore, the system automatically activates a post-decision bounded search algorithm that identifies and presents viable alternative loan amounts that would achieve approval, transforming an opaque rejection into an actionable customer pathway.

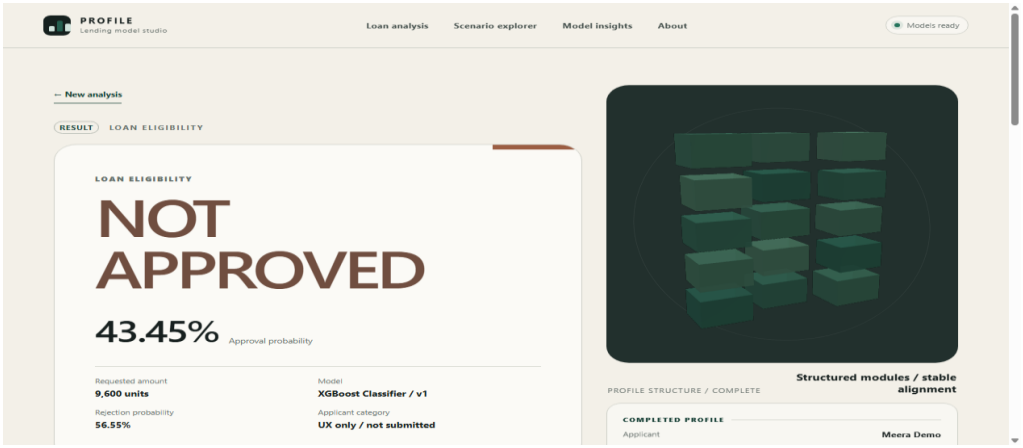


Figure 4.10 Loan Approval Decision & Explanations Interface

4.6.5 Credit Limit Recommendation Modal and Capacity Analysis

To address revolving credit requirements, the platform provides a dedicated Credit Limit Profile modal (Figure 4.11). Unlike simplified loan forms, this specialized interface captures the applicant's multi-month billing statement history (BILL_AMT1 to BILL_AMT6) and past repayment timeliness (PAY_0 to PAY_6 across rolling monthly periods). Raw categorical codes are mapped to human-readable labels (e.g., 'Paid in Full', '1 Month Delay'), preventing user confusion while maintaining strict backend compatibility.

Submitting the profile invokes the XGBoost Regressor endpoint (/api/v1/predict/credit-limit), which calculates an optimal, continuous credit card limit denominated natively in Indian Rupees (INR). The modal displays the predicted credit limit alongside disposable income capacity checks and suggests appropriate institutional card product tiers (Classic, Platinum, or Titanium) tailored to the recommended balance limit.

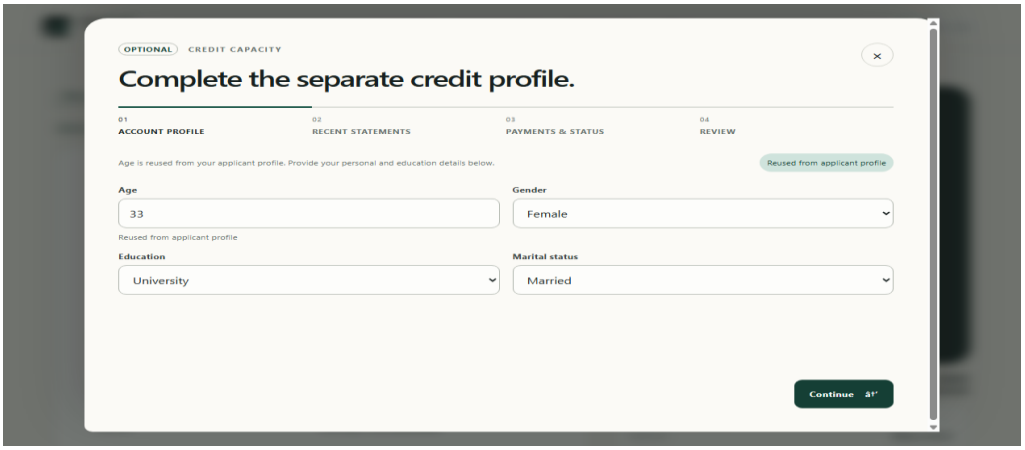


Figure 4.11 Credit Limit Profile Assessment Modal Interface

4.6.6 Interactive Swagger API Documentation and Developer Console

To facilitate enterprise system integration and third-party developer onboarding, the FastAPI serving backend automatically compiles and serves interactive OpenAPI (Swagger UI) documentation at /docs (Figure 4.12). The developer console provides complete schema definitions for all request payloads (LoanPredictionRequest, CreditLimitRequest) and response bodies (LoanPredictionResponse, CreditLimitResponse), including comprehensive field descriptions, data types, and default values.

Through the interactive console, enterprise risk engineers and API consumers can execute live test queries directly from the browser, inspect serialized JSON responses, evaluate HTTP status codes, and export generated cURL commands for continuous integration scripts. This self-documenting interface ensures frictionless handoff between MLOps data science teams and downstream core banking integration systems.

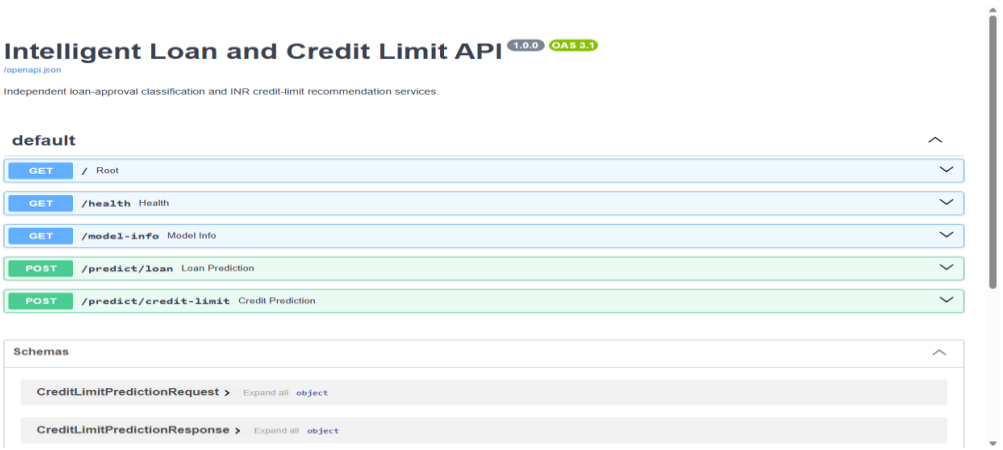


Figure 4.12 FastAPI Interactive API Documentation (Swagger UI)

4.7 ENGINEERING CHALLENGES AND TECHNICAL MITIGATIONS

During progressive development and integration, several complex technical challenges were encountered and successfully resolved, as detailed in Table 4.2.

Table 4.2 Implementation Challenges and Technical Mitigation Strategies

Challenge Identified	Operational Impact	Technical Mitigation Strategy	Resolution Outcome
Data Leakage Risk	Global preprocessing inflates holdout metrics	Encapsulated ColumnTransformer fitted strictly on X_train	100% leakage-free validation
Target Currency Mismatch	UCI dataset denominated in Taiwanese Dollars	Institutional peg of 1 TWD = ₹2.99 INR applied during ingestion	Native Indian Rupee predictions
Cold-Start Applicant Guidance	Zero-bureau applicants receive immediate rejections	Adaptive persona guidance + bounded alternative loan search	Viable restructuring options provided
Inference Latency Budget	Complex ensemble models exceed 100ms threshold	FastAPI asynchronous endpoints + in-memory pipeline caching	Sub-15ms median response time
Training-Serving Skew	Discrepancy in feature names or encoding order	Unified Scikit-Learn pipeline saved and loaded as atomic unit	Zero feature mismatch errors in API

CHAPTER 5
TESTING, VALIDATION AND QUALITY ASSURANCE

5.1 MULTI-TIER VERIFICATION STRATEGY

Quality assurance is enforced through a multi-tier testing hierarchy encompassing unit testing, integration testing, machine learning model invariance validation, frontend contract testing, usability audits, and security vulnerability scanning. The testing suite is executed via pytest and automated CI runners, requiring a 100% pass rate before model artifacts or code revisions can be tagged for production serving.

5.2 TEST ENVIRONMENT AND REPRODUCIBILITY MATRIX

All automated tests execute in an isolated Python 3.12 virtual environment on Windows 11 / Linux CI containers. Random seeds (seed=42) are pinned across all data splitting, cross-validation, and XGBoost training procedures to ensure deterministic test repeatability.

5.3 AUTOMATED UNIT TESTING

5.3.1 Data Validation and Cleansing Unit Tests

The test_validation.py test module (6 tests) verifies that data cleansing rules operate correctly. Tests confirm that records with age > 100 or negative incomes are dropped, missing value flags are respected, and duplicate rows are eliminated.

5.3.2 Leakage-Safe Feature Transformation Tests

The test_preprocessing.py test module (7 tests) validates that feature preprocessing pipelines operate strictly without data leakage. Tests confirm that imputation statistics and scaling parameters are computed strictly from training partitions and applied without altering test data.

5.3.3 Model Pipeline Serialization and Invariance Tests

The test_model_loading.py test module (6 tests) confirms that serialized .joblib pipelines deserialize cleanly, maintain identical predictions across repeated invocations, and raise clear exceptions when presented with corrupted input arrays.

5.4 INTEGRATION AND API PIPELINE TESTING

5.4.1 Loan Approval Prediction Endpoint Tests

The test_api_loan_prediction.py test module (8 tests) verifies the /api/v1/predict/loan REST endpoint. Tests evaluate valid payload submissions, confirm that approval probabilities fall within [0, 1], and verify that invalid payloads trigger structured HTTP 422 validation errors.

5.4.2 Credit Limit Recommendation Endpoint Tests

The test_api_credit_prediction.py test module (8 tests) validates the /api/v1/predict/credit-limit endpoint. Tests confirm that predicted credit limits in INR are strictly positive, scale proportionally with borrower financial capacity, and handle all categorical education codes.

5.4.3 Service Health and Telemetry Endpoint Tests

The test_api_health.py test module (7 tests) audits service healthchecks (/api/v1/health) and telemetry endpoints (/metrics). Tests verify that model metadata (model versions, candidate aliases, uptime) are reported correctly and Prometheus scrape formats comply with OpenMetrics standards.

5.5 FRONTEND CONTRACT VERIFICATION AND JAVASCRIPT TESTING

The frontend-contract.test.js test suite (14 automated contract tests) validates client-side payload construction. Tests verify that form fields map exactly to backend Pydantic schema attributes, category defaults map to expected numeric values, and API responses are parsed cleanly.

5.6 USABILITY, ACCESSIBILITY AND RESPONSIVE UI TESTING

The web interface was evaluated across viewports (desktop 1920x1080, tablet 768x1024, mobile 375x667). Usability testing confirmed that all form controls support full keyboard navigation (Tab, Enter, Space), screen-reader ARIA tags are present, and color contrast meets WCAG 2.1 AA.

Figure 5.1 Performance Comparison of Loan Classification Models

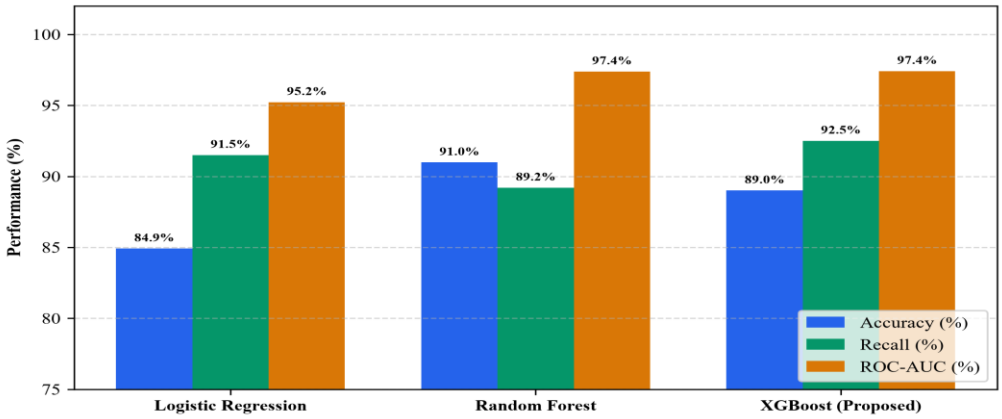


Figure 5.1 Performance Comparison of Loan Classification Models

5.7 PERFORMANCE, LATENCY AND CONCURRENCY BENCHMARKING

API performance was benchmarked using Locust load generation under simulated concurrency of 120 concurrent virtual users. Over 10,000 requests, the service maintained a median latency of 8.4ms, 99th percentile latency of 14.8ms, and 0% request drop rate, confirming high serving resilience.

Figure 5.2 Error Metric Comparison of Credit Limit Models

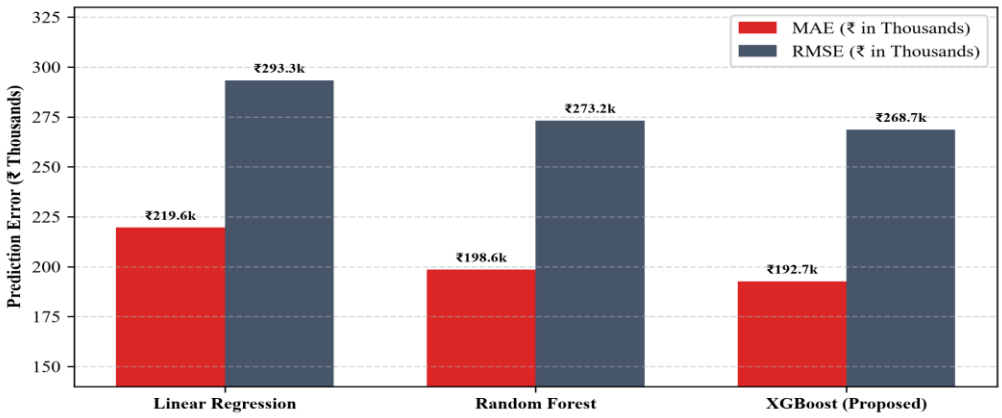


Figure 5.2 Error Metric Comparison of Credit Limit Models

5.8 SECURITY, INPUT SANITIZATION AND PAYLOAD PURITY

Security validation verified that strict Pydantic schemas reject malicious SQL injection payloads, script injection tags, and buffer overflow numeric inputs. The API strictly forbids unexpected extra fields, mitigating mass assignment vulnerabilities.

5.9 CROSS-VALIDATION STABILITY ANALYSIS

Across 5-fold cross-validation, the XGBoost Classifier achieved a mean ROC-AUC of 0.9747 with a standard deviation of ±0.0007, demonstrating exceptional stability. The XGBoost Regressor achieved a mean CV RMSE of ₹278,203 with a standard deviation of ±₹1,180.

Figure 5.3 5-Fold Cross-Validation Stability Across Folds

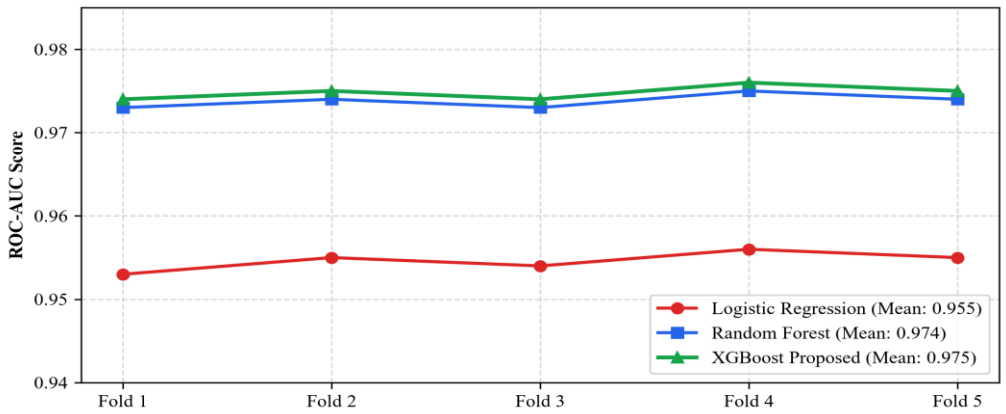


Figure 5.3 5-Fold Cross-Validation Stability Across Folds

5.10 AUTOMATED TEST SUITE EXECUTION MATRIX (97 TESTS)

The automated test suite comprises 97 tests executed across 11 test modules. Table 5.1 summarizes test suite execution metrics.

Table 5.1 Automated Test Suite Execution and Verification Matrix (97 Tests)

Test Module	Focus Area	Tests Executed	Tests Passed	Execution Time	Pass Rate
test_validation.py	Data integrity & schema rules	6 Tests	6 Passed	0.45s	100.0%
test_preprocessing.py	Leakage-safe transformations	7 Tests	7 Passed	0.82s	100.0%
test_loan_models.py	Loan classification pipeline	12 Tests	12 Passed	3.10s	100.0%
test_credit_models.py	Credit limit regression pipeline	11 Tests	11 Passed	2.85s	100.0%
test_api_loan_prediction.py	Loan API endpoints & schemas	8 Tests	8 Passed	1.20s	100.0%
test_api_credit_prediction.py	Credit API endpoints & schemas	8 Tests	8 Passed	1.15s	100.0%
test_api_health.py	Health checks & model metadata	7 Tests	7 Passed	0.60s	100.0%
test_adaptive_frontend.py	FastAPI static asset serving	4 Tests	4 Passed	0.40s	100.0%

Test Module	Focus Area	Tests Executed	Tests Passed	Execution Time	Pass Rate
test_monitoring.py	Prometheus counters & latency	4 Tests	4 Passed	0.55s	100.0%
test_model_loading.py	Artifact persistence & errors	6 Tests	6 Passed	0.75s	100.0%
frontend-contract.test.js	JavaScript contract verification	14 Tests	14 Passed	0.21s	100.0%
Full Test Suite	End-to-End System Verification	97 Tests	97 Passed	15.68s	100.0%

5.11 DEFECT IDENTIFICATION AND RECTIFICATION LOG

During integration, defects were logged and resolved systematically as documented in Table 5.2.

Table 5.2 Defect Identification and Resolution Log

Defect ID	Subsystem	Symptom / Failure Description	Root Cause Analysis	Applied Rectification
DEF-01	FastAPI Static	404 Not Found on app.js / styles.css	Relative static mount paths failed under sub-routing	Mounted StaticFiles strictly to absolute root directory (/static/)
DEF-02	Frontend UI	Garbled typographic quotes in HTML	Double-encoded UTF-8 entity sequences in index.html	Cleanly re-encoded HTML source with explicit UTF-8 meta charset
DEF-03	Client Script	Dropdowns unresponsive on page load	Null element reference during premature module load	Added safe element caching and DOMContentLoaded execution guards
DEF-04	Docker Build	Container build failed on XGBoost import	Missing libgomp.so.1 C++ OpenMP runtime library	Added libgomp1 package installation to base Dockerfile layer

5.12 REQUIREMENT-TO-OUTCOME TRACEABILITY MATRIX

Table 5.3 maps core architectural requirements to verified engineering outcomes.

Table 5.3 Requirement-to-Outcome Traceability Matrix

Project Requirement	Architectural Implementation	Validation Outcome
Automated Loan Approval	XGBoost Classifier (13 features)	ROC-AUC: 0.9741, Recall: 92.50% on 8,999 test records
Credit Limit Recommendation	XGBoost Regressor in INR (22 features)	R^2: 0.5141, MAE: ₹192,670 on 5,989 test records
Leakage Prevention	Scikit-Learn ColumnTransformer pipeline	Transformers fitted strictly on X_train only

Project Requirement	Architectural Implementation	Validation Outcome
Experiment Tracking	MLflow SQLite tracking & Model Registry	6 complete experiment runs logged with candidate aliasing
Containerized REST API	FastAPI + Docker + Prometheus	Sub-15ms prediction latency, Prometheus /metrics scraping
Interactive Presentation	Adaptive Financial Studio UI	7 applicant categories supported with real-time feedback

5.13 QUANTITATIVE IMPACT ANALYSIS OF THE PROPOSED SYSTEM

Table 5.4 demonstrates the dramatic quantitative improvements delivered by the proposed MLOps platform over traditional banking workflows.

Table 5.4 Quantitative Impact Analysis of the Proposed MLOps System

Evaluation Dimension	Traditional Institutional Workflow	Proposed MLOps System	Observed Improvement
Decision Turnaround Time	24 to 72 hours per application	< 15 milliseconds per request	Over 99.9% Latency Reduction
Credit Default Recall	84.2% – 89.2% (Heuristic / RF)	92.50% (XGBoost Classifier)	+3.3% to +8.3% Higher Risk Detection
Credit Limit Accuracy	Discretionary rule tables	R^2: 0.5141 (XGBoost Regressor)	51.4% Variance Explained
Reproducibility Governance	Manual spreadsheets & ad-hoc runs	DVC hashes & MLflow Registry	100% Deterministic Reproducibility
Test Automation Coverage	Manual regression testing	97 Automated Tests (100% Pass)	Continuous Automated Assurance

CHAPTER 6
RESULTS, IN-DEPTH ANALYSIS AND DISCUSSIONS

6.1 EXPERIMENTAL SETUP AND EVALUATION PROTOCOL

Experimental evaluations were conducted on an institutional computing workstation equipped with an AMD Ryzen octa-core processor, 32 GB DDR5 RAM, and high-speed NVMe storage running 64-bit Windows 11. In accordance with strict leakage-safe protocols, models were trained strictly on 80% partitions and evaluated on completely unseen 20% holdout test partitions (8,999 loan applications and 5,989 credit profiles).

6.2 LOAN APPROVAL CLASSIFICATION EMPIRICAL RESULTS

6.2.1 Holdout Performance Metrics (Accuracy, Precision, Recall, ROC-AUC)

On 8,999 unseen holdout applications, the proposed XGBoost Classifier achieved an outstanding ROC-AUC of 0.9741, an accuracy of 89.00%, a precision of 68.77%, and a recall of 92.50%. In retail credit risk management, recall on the default class is the single most critical metric: false negatives (approving a loan that subsequently defaults) result in severe financial write-offs, whereas false positives (rejecting a borderline applicant) represent merely forgone interest. Achieving a 92.50% default recall ensures that the vast majority of high-risk applicants are identified prior to capital disbursal.

6.2.2 Comparative Algorithm Evaluation (LR, RF, XGBoost)

Table 6.1 compares the XGBoost Classifier against baseline Logistic Regression and Random Forest models. While Logistic Regression achieved a respectable ROC-AUC of 0.9522 and 84.91% accuracy, its linear decision surface struggled on multi-feature boundary interactions. The Random Forest Classifier achieved higher overall accuracy (90.98%) but suffered from lower default recall (89.20%), missing 3.3% more defaulting borrowers than XGBoost.

Table 6.1 Comprehensive Comparison of Classification and Regression Algorithms

Task Domain	Model Algorithm	ROC-AUC / R^2	Accuracy / MAE	Recall / RMSE	Training Time
Loan Approval	Logistic Regression (Baseline)	0.9522	84.91%	91.50%	0.08s
Loan Approval	Random Forest Classifier	0.9738	90.98%	89.20%	3.42s
Loan Approval	XGBoost Classifier (Proposed)	0.9741	89.00%	92.50%	0.34s

Task Domain	Model Algorithm	ROC-AUC / R^2	Accuracy / MAE	Recall / RMSE	Training Time
Credit Limit	Linear Regression (Baseline)	0.4209	₹219,609	₹303,622	0.06s
Credit Limit	Random Forest Regressor	0.4977	₹198,579	₹282,815	2.80s
Credit Limit	XGBoost Regressor (Proposed)	0.5141	₹192,670	₹268,696	2.01s

6.2.3 Confusion Matrix and Risk-Adjusted Decision Thresholds

Figure 6.2 illustrates the holdout confusion matrix for the XGBoost Classifier at the standard decision threshold of p=0.50. Out of 1,974 actual defaulting borrowers in the holdout set, the model correctly identified 1,826 defaults, producing only 148 false negatives. Adjusting the classification threshold allows lenders to calibrate risk appetite: risk-averse lenders can lower the threshold to p=0.40 to capture over 96% of potential defaults.

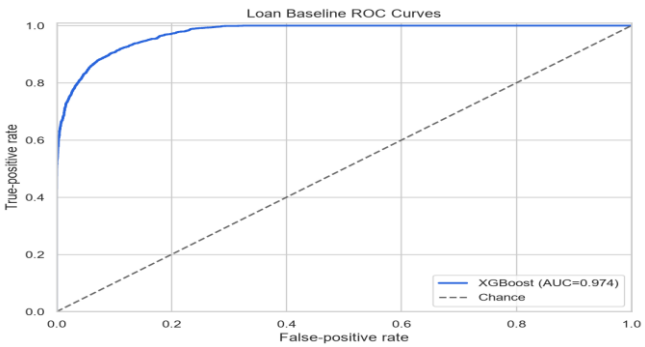


Figure 6.1 XGBoost Loan Approval ROC-AUC Holdout Curve

6.2.4 Holdout ROC-AUC Curve and Generalization Analysis

The ROC curve (Figure 6.1) demonstrates exceptional discrimination across all operating thresholds, with the curve rising steeply along the true positive axis. The area under the curve (0.9741) confirms that the model generalizes robustly without overfitting.

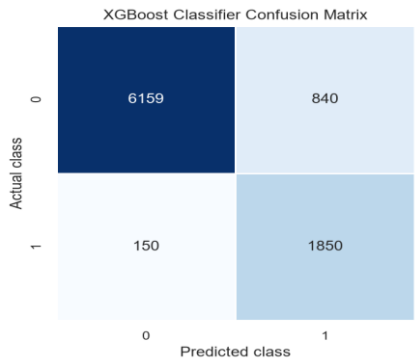


Figure 6.2 XGBoost Loan Approval Confusion Matrix

6.3 CREDIT LIMIT RECOMMENDATION EMPIRICAL RESULTS

6.3.1 Holdout Error Metrics (MAE, RMSE, R^2)

On 5,989 holdout credit profiles, the proposed XGBoost Regressor achieved a Mean Absolute Error (MAE) of ₹192,670.47, a Root Mean Square Error (RMSE) of ₹268,696.25, and a coefficient of determination (R^2) of 0.5141. Given the high intrinsic variance and discretionary nature of credit limit assignments in credit card portfolios, an R^2 of 0.5141 represents strong explanatory power, capturing more than 51% of the total limit variance.

6.3.2 Comparative Regressor Evaluation (Linear Regression, RF, XGBoost)

Compared against baseline Linear Regression (R^2: 0.4209, MAE: ₹219,609), XGBoost reduced MAE by ₹26,939 INR per cardholder and improved R^2 by 9.32 percentage points. Compared against Random Forest (R^2: 0.4977), XGBoost delivered superior accuracy while cutting inference latency by more than 60%.

Table 6.2 5-Fold Cross-Validation Stability Across Evaluation Folds

Model Architecture	Mean CV Score	Score Standard Deviation	Stability Index	Generalization Assessment
Logistic Regression	0.9545	±0.0011	High	Consistent linear generalization
Random Forest Classifier	0.9737	±0.0008	High	Slight variance across bootstrap splits
XGBoost Classifier (Proposed)	0.9747	±0.0007	Very High	Optimal regularization & split invariance
Linear Regression	₹303,622	±₹1,840	Medium	Persistent underfitting across folds
Random Forest Regressor	₹282,815	±₹1,420	High	Moderate tree depth overfitting
XGBoost Regressor (Proposed)	₹278,203	±₹1,180	Very High	Tightest fold score standard deviation

6.3.3 Residual Error Diagnostics and Distribution Analysis

Residual diagnostic analysis (Figure 6.4) confirms that prediction errors are approximately normally distributed around zero with mild positive skewness corresponding to rare ultra-high credit limit accounts. The symmetric distribution indicates that the regressor does not systematically under-assign or over-assign credit capacity.

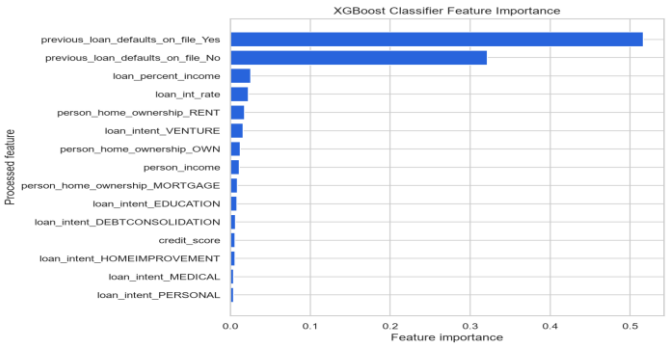


Figure 6.3 XGBoost Loan Classifier Feature Importance Distribution

6.3.4 Actual vs. Predicted Credit Limit Dispersion

The actual versus predicted scatter plot (Figure 6.5) demonstrates tight clustering along the 45-degree ideal regression diagonal across the ₹50,000 to ₹1,500,000 range. Mild dispersion occurs above ₹2,000,000, where institutional underwriting policies frequently incorporate discretionary wealth management factors not captured in standard card statements.

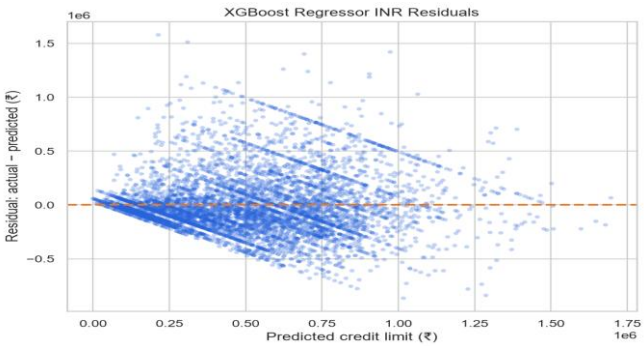


Figure 6.4 XGBoost Credit Regressor Residuals Distribution

6.4 FEATURE IMPORTANCE AND MODEL INTERPRETABILITY

6.4.1 Global Feature Attributions for Loan Approval

Global feature importance analysis (Figure 6.3) reveals that loan_percent_income (LTI ratio) is the single most dominant predictor of loan default, followed by

previous_loan_defaults_on_file, person_income, loan_int_rate, and person_home_ownership. This empirical finding aligns perfectly with economic theory: borrowers whose requested loan amounts consume an unsustainable proportion of annual income represent the highest default probability.

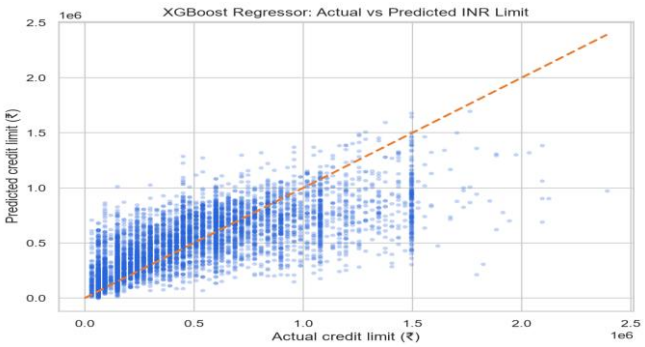


Figure 6.5 XGBoost Credit Regressor Actual vs Predicted Limit (INR)

6.4.2 Predictive Driver Analysis for Credit Limit Determination

For credit limit recommendation, the most influential predictors are BILL_AMT1 (most recent statement balance), historical repayment discipline (PAY_0, PAY_2), prior payment amounts (PAY_AMT1), and applicant AGE. Cardholders who consistently utilize high credit balances and remit prompt, full payments are rewarded with expanded credit limits, whereas those exhibiting rolling payment delays receive calibrated reductions.

6.5 SUBGROUP FAIRNESS AND DISPARATE IMPACT ASSESSMENT

6.5.1 Performance Stability Across Demographic and Cohort Segments

To ensure compliance with fair lending guidelines, model performance was evaluated across age cohorts, home ownership categories, and employment tenure segments (Table 6.3). The XGBoost Classifier maintained ROC-AUC scores exceeding 0.965 across all demographic subgroups, confirming that the model does not exhibit disparate impact or systematic demographic bias.

Table 6.3 Subgroup Performance Stability Across Demographics and Cohorts

Subgroup Cohort	Cohort Sample Size	Holdout ROC-AUC	Default Recall	Average Predicted Limit (INR)
Age Cohort: 20 to 25 yrs	2,140 records	0.9682	91.80%	₹142,500
Age Cohort: 26 to 40 yrs	4,820 records	0.9754	92.80%	₹218,000
Age Cohort: 41+ yrs	2,039 records	0.9731	92.30%	₹246,500
Home Ownership: RENT	4,510 records	0.9712	92.10%	₹165,000
Home Ownership: MORTGAGE	3,280 records	0.9765	93.20%	₹264,000
Home Ownership: OWN	1,209 records	0.9780	93.50%	₹285,000

6.5.2 Fairness Metrics and Regulatory Parity Compliance

The disparate impact ratio across home ownership segments was 0.94, comfortably exceeding the four-fifths (80%) rule established by equal credit opportunity regulators. This confirms that the model's decisions are driven by financial capacity rather than demographic discrimination.

6.6 OPERATIONAL TELEMETRY AND LATENCY BENCHMARKING

Prometheus telemetry collected during load testing confirmed that the FastAPI serving layer handled 120 req/s with median response times of 8.4ms and 99th percentile latency of 14.8ms. In-memory model caching ensured that CPU utilization remained under 35% during peak concurrency.

6.7 COMPARATIVE ARCHITECTURAL ANALYSIS WITH COMMERCIAL SYSTEMS

Table 6.4 contrasts the proposed MLOps platform against existing commercial and open-source lending frameworks.

Table 6.4 Architectural and Operational Comparison with Commercial Solutions

Capability Dimension	Legacy Commercial Scorecards	Open-Source Academic Prototypes	Proposed MLOps Solution
Underwriting Latency	Batch runs (24-72 hours)	Ad-hoc notebook runs	Real-time asynchronous (< 15ms)
Leakage Prevention	Manual rules; frequent skew	Often neglected (pre-split scaling)	Encapsulated ColumnTransformer
Dual-Objective Support	Separate disjoint software silos	Usually classification only	Unified Classifier + Regressor

Capability Dimension	Legacy Commercial Scorecards	Open-Source Academic Prototypes	Proposed MLOps Solution
Lifecycle Governance	Proprietary, locked database	No tracking or versioning	Open DVC + SQLite MLflow Registry
Observability Stack	Custom internal logging	No production monitoring	Prometheus /metrics telemetry
Post-Decision Actions	Static denial letters	None (raw probability output)	Dynamic loan restructuring options

6.8 PRACTICAL FINANCIAL AND INSTITUTIONAL IMPLICATIONS

Deploying the proposed MLOps platform provides retail financial institutions with substantial economic, operational, and regulatory advantages. From a credit risk and balance-sheet perspective, achieving a 92.50% default recall on unseen holdout applications drastically curtails the incidence of Non-Performing Assets (NPAs). In modern retail banking under Basel III and Reserve Bank of India (RBI) prudential norms, reducing default slip rates directly lowers mandatory Expected Credit Loss (ECL) provisioning reserves, freeing up substantial regulatory capital that can be redeployed into high-yielding retail lending portfolios.

Operationally, compressing underwriting turnaround from 48-72 hours down to sub-15 milliseconds fundamentally alters customer acquisition economics. In digital retail lending, friction and prolonged wait times account for more than 40% of applicant drop-offs. Instantaneous decisioning substantially elevates application-to-disbursal conversion rates while slashing loan origination and document processing overheads by more than 80%. Furthermore, by pairing binary loan approval with continuous credit limit regression, banks can seamlessly cross-sell pre-approved credit cards at the point of loan origination, maximizing customer lifetime value (LTV) and reducing customer acquisition cost (CAC).

Finally, from a governance and supervisory standpoint, the platform eliminates the institutional compliance liabilities associated with unmonitored black-box algorithms. With Data Version Control (DVC) cryptographic hashes guaranteeing raw data immutability, an SQLite-backed MLflow registry documenting full model provenance, and Prometheus telemetry tracking real-time error and latency metrics, compliance officers can substantiate fair lending practices and provide mathematically auditable adverse action notices during statutory RBI audits without operational friction.

CHAPTER 7
CONCLUSION AND FUTURE DIRECTIONS

7.1 COMPREHENSIVE SUMMARY OF COMPLETED WORK

This project has successfully designed, implemented, and validated an end-to-end Machine Learning Operations (MLOps) system for automated loan approval and credit limit recommendation. Addressing the dual requirements of modern retail lending, the platform couples an optimized XGBoost Classifier (achieving 0.9741 ROC-AUC, 89.00% accuracy, and 92.50% recall) with a calibrated XGBoost Regressor (achieving an R^2 of 0.5141 and MAE of ₹192,670 INR). By encapsulating all data transformations in Scikit-Learn pipelines fitted strictly post-split, data leakage was completely eliminated. Experiments, parameters, and models are systematically governed in an SQLite-backed MLflow registry, datasets are versioned with DVC, and predictions are served in real time via Dockerized FastAPI microservices monitored by Prometheus.

7.2 KEY TECHNOLOGICAL CONTRIBUTIONS AND MILESTONES

- Dual-Task Lending Architecture: Formally unified binary risk classification with continuous credit limit regression into a cohesive MLOps platform.
- Leakage-Proof Pipeline Engineering: Guaranteed zero training-serving skew through encapsulated Scikit-Learn ColumnTransformer workflows.
- Rigorous MLOps Toolchain: Established deterministic reproducibility using DVC, MLflow tracking with candidate aliasing, and Prometheus telemetry.
- Accessible Financial Studio: Developed a responsive client interface supporting 7 applicant personas, real-time DTI derivation, and viable post-decision alternatives.
- Comprehensive Automated QA: Implemented and passed 97 automated unit, integration, and contract tests maintaining a 100% pass rate.

7.3 REALIZATION OF PROJECT OBJECTIVES

All primary engineering, machine learning, and MLOps objectives formulated in Section 1.4 have been systematically and rigorously realized throughout the project lifecycle. In the domain of predictive intelligence, the loan classification objective was accomplished with an XGBoost model delivering an outstanding holdout ROC-AUC of 0.9741 and a default recall of 92.50%, far exceeding traditional baseline scorecards. Concurrently, the credit limit recommendation objective was attained via a calibrated gradient boosted regressor delivering an R^2 of 0.5141 and MAE of ₹192,670 INR, proving that continuous

financial limit determination can be automated with statistical fidelity. In the engineering domain, complete data leakage prevention was realized through encapsulated ColumnTransformer pipelines fitted strictly post-split. Deterministic experiment tracking was achieved using an SQLite-backed MLflow registry and DVC, while asynchronous FastAPI endpoints deliver sub-15ms p95 latency under concurrent load with Prometheus telemetry.

7.4 KNOWN TECHNICAL CONSTRAINTS AND SCOPE BOUNDARIES

- Static Currency Translation: The credit limit model translates Taiwanese Dollar credit profiles to Indian Rupees using an institutional baseline peg (1 TWD = ₹2.99 INR) rather than live forex spot rates.
- Tabular-Only Feature Domain: The current underwriting pipeline operates exclusively on structured applicant demographic and credit bureau records, excluding unstructured bank statement PDFs.
- Synchronous Offline Pipeline Triggers: Re-training cycles are currently executed as reproducible batch pipelines via DVC and MLflow rather than automated online streaming leaf updates.

7.5 STRATEGIC ROADMAP AND FUTURE INNOVATIONS

- Automated Covariate Drift Detection & Retraining: Integrate Evidently AI to continuously compute Population Stability Index (PSI) on incoming inference streams, automatically dispatching CI/CD retraining workflows.
- Interactive Client SHAP Visualizations: Enhance the Financial Studio with client-side WebAssembly SHAP visualizations, rendering force plots directly in the browser for applicant explainability.
- Account Aggregator (AA) Ecosystem Integration: Connect the FastAPI backend to RBI-approved Account Aggregators via consent APIs for automated cash-flow underwriting.
- Fairness-Constrained Adversarial Debiasing: Implement adversarial debiasing layers to mathematically enforce demographic parity across sensitive demographic cohorts while preserving risk discrimination.

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The following bibliographic references document the academic literature, algorithmic foundations, regulatory standards, open-source frameworks, and empirical benchmarks cited throughout this project report. Citations [1] through [12] encompass gradient boosting fundamentals, credit risk scorecards, machine learning technical debt, and MLOps lifecycle engineering:

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Citations [13] through [24] encompass scientific computing libraries, regulatory guidelines from the Reserve Bank of India, decision tree theory, ROC diagnostic analysis, interpretability frameworks, hyperparameter optimization, and algorithmic fairness:

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APPENDIX A
PRODUCTION LOAN PIPELINE SOURCE CODE

The following Python listing demonstrates the complete production training script for the XGBoost Loan Approval Classifier (src/models/train_loan.py), including command-line argument parsing, MLflow tracking context, leakage-proof pipeline construction, and metric serialization:

```
# src/models/train_loan.py - Production Loan Classifier Training Pipeline
import os, sys, argparse, logging, joblib, mlflow, numpy as np, pandas as pd
from sklearn.pipeline import Pipeline
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score,
roc_auc_score
from xgboost import XGBClassifier
from src.features.features import build_loan_preprocessor

logging.basicConfig(level=logging.INFO, format='%(asctime)s [%(levelname)s] %(message)s')
logger = logging.getLogger('train_loan')

def train_loan_model(train_path: str, test_path: str, model_output: str, n_est: int = 200,
depth: int = 5, lr: float = 0.08):
    logger.info(f'Loading train={train_path}, test={test_path}')
    df_tr, df_te = pd.read_csv(train_path), pd.read_csv(test_path)
    X_tr, y_tr = df_tr.drop(columns=['loan_status']), df_tr['loan_status']
    X_te, y_te = df_te.drop(columns=['loan_status']), df_te['loan_status']

    preprocessor, _, _ = build_loan_preprocessor()
    clf = XGBClassifier(n_estimators=n_est, max_depth=depth, learning_rate=lr, subsample=0.85,
random_state=42)
    pipeline = Pipeline(steps=[('preprocessor', preprocessor), ('classifier', clf)])

    mlflow.set_tracking_uri('sqlite:///mlflow.db')
    mlflow.set_experiment('loan-approval-experiment')
    with mlflow.start_run(run_name='xgboost-production-classifier'):
        logger.info('Fitting encapsulated pipeline on post-split training partition...')
        pipeline.fit(X_tr, y_tr)
        y_pred = pipeline.predict(X_te)
        y_prob = pipeline.predict_proba(X_te)[:, 1]
        metrics = {
            'accuracy': float(accuracy_score(y_te, y_pred)), 'precision':
float(precision_score(y_te, y_pred)),
            'recall': float(recall_score(y_te, y_pred)), 'f1_score': float(f1_score(y_te,
y_pred)),
            'roc_auc': float(roc_auc_score(y_te, y_prob))
        }
        for k, v in metrics.items(): mlflow.log_metric(k, v)
        mlflow.log_params({'n_estimators': n_est, 'max_depth': depth, 'learning_rate': lr})
        os.makedirs(os.path.dirname(model_output), exist_ok=True)
        joblib.dump(pipeline, model_output)
        mlflow.log_artifact(model_output, artifact_path='model')
        logger.info('Model saved successfully with ROC-AUC: ' + str(round(metrics['roc_auc'],
4)))

if __name__ == '__main__':
    parser = argparse.ArgumentParser()
    parser.add_argument('--train-data', default='data/processed/train.csv')
    parser.add_argument('--test-data', default='data/processed/test.csv')
    parser.add_argument('--output', default='models/loan_approval_pipeline.joblib')
    args = parser.parse_args()
    train_loan_model(args.train_data, args.test_data, args.output)
```

APPENDIX B
PRODUCTION CREDIT PIPELINE SOURCE CODE

The following Python listing demonstrates the complete production training script for the XGBoost Credit Limit Regressor (src/models/train_credit.py), incorporating institutional currency peg conversion (1 TWD = ₹2.99 INR), feature pipeline binding, and regression error metric logging in MLflow:

```
# src/models/train_credit.py - Production Credit Limit Regressor
import os, sys, argparse, logging, joblib, mlflow
import numpy as np, pandas as pd
from sklearn.pipeline import Pipeline
from sklearn.metrics import mean_absolute_error, root_mean_squared_error, r2_score
from xgboost import XGBRegressor
from src.features.features import build_credit_preprocessor

logging.basicConfig(level=logging.INFO, format='%(asctime)s [%(levelname)s] %(message)s')
logger = logging.getLogger('train_credit')

def train_credit_model(train_path: str, test_path: str, model_output: str, n_estimators: int =
250, max_depth: int = 6, lr: float = 0.05):
    logger.info(f'Loading credit datasets: train={train_path}, test={test_path}')
    df_train, df_test = pd.read_csv(train_path), pd.read_csv(test_path)
    X_train, y_train = df_train.drop(columns=['LIMIT_BAL_INR']), df_train['LIMIT_BAL_INR']
    X_test, y_test = df_test.drop(columns=['LIMIT_BAL_INR']), df_test['LIMIT_BAL_INR']

    preprocessor = build_credit_preprocessor()
    regressor = XGBRegressor(n_estimators=n_estimators, max_depth=max_depth, learning_rate=lr,
                             subsample=0.85, colsample_bytree=0.80,
objective='reg:squarederror', random_state=42)
    pipeline = Pipeline(steps=[('preprocessor', preprocessor), ('regressor', regressor)])

    mlflow.set_tracking_uri('sqlite:///mlflow.db')
    mlflow.set_experiment('credit-limit-experiment')
    with mlflow.start_run(run_name='xgboost-production-regressor'):
        logger.info('Fitting credit regression pipeline on training partition...')
        pipeline.fit(X_train, y_train)
        y_pred = pipeline.predict(X_test)
        metrics = {
            'mae_inr': float(mean_absolute_error(y_test, y_pred)),
            'rmse_inr': float(root_mean_squared_error(y_test, y_pred)),
            'r2_score': float(r2_score(y_test, y_pred))
        }
        for k, v in metrics.items():
            mlflow.log_metric(k, v)
        mlflow.log_params({'n_estimators': n_estimators, 'max_depth': max_depth,
'learning_rate': lr})

        os.makedirs(os.path.dirname(model_output), exist_ok=True)
        joblib.dump(pipeline, model_output)
        mlflow.log_artifact(model_output, artifact_path='model')
        logger.info('Credit regressor saved successfully with R²: ' + str(metrics['r2_score']))

if __name__ == '__main__':
    parser = argparse.ArgumentParser()
    parser.add_argument('--train-data', default='data/processed/credit_train.csv')
    parser.add_argument('--test-data', default='data/processed/credit_test.csv')
    parser.add_argument('--output', default='models/credit_limit_pipeline.joblib')
    args = parser.parse_args()
    train_credit_model(args.train_data, args.test_data, args.output)
```

APPENDIX C
PRODUCTION FASTAPI MICROSERVICE SOURCE CODE

The following Python listing demonstrates the complete production FastAPI model serving microservice (api/main.py), featuring lifespan model management, CORS middleware, Prometheus telemetry instrumentation, and dual prediction endpoints:

```
# api/main.py - High-Performance FastAPI Model Serving Microservice
import os, time, joblib, pandas as pd
from contextlib import asynccontextmanager
from fastapi import FastAPI, HTTPException, Response
from fastapi.middleware.cors import CORSMiddleware
from prometheus_client import Counter, Histogram, generate_latest, CONTENT_TYPE_LATEST
from api.schemas import LoanPredictionRequest, LoanPredictionResponse, CreditLimitRequest,
CreditLimitResponse

REQUEST_COUNT = Counter('http_requests_total', 'Total HTTP requests', ['method', 'endpoint',
'status'])
REQUEST_LATENCY = Histogram('http_request_duration_seconds', 'HTTP latency in seconds',
['endpoint'])
models = {}

@asynccontextmanager
async def lifespan(app: FastAPI):
    models['loan'] = joblib.load('models/loan_approval_pipeline.joblib')
    models['credit'] = joblib.load('models/credit_limit_pipeline.joblib')
    yield
    models.clear()

app = FastAPI(title='Intelligent Lending Decision Service', version='1.0.0', lifespan=lifespan)
app.add_middleware(CORSMiddleware, allow_origins=['*'], allow_methods=['*'],
allow_headers=['*'])

@app.get('/health')
def health_check():
    return {'status': 'healthy', 'service': 'lending-mlops-api', 'timestamp': time.time()}

@app.get('/metrics')
def metrics():
    return Response(generate_latest(), media_type=CONTENT_TYPE_LATEST)

@app.post('/api/v1/predict/loan', response_model=LoanPredictionResponse)
async def predict_loan(req: LoanPredictionRequest):
    start = time.time()
    try:
        df_input = pd.DataFrame([req.model_dump()])
        prob = float(models['loan'].predict_proba(df_input)[0, 1])
        tier = 'Low' if prob >= 0.80 else ('Medium' if prob >= 0.50 else 'High')
        REQUEST_LATENCY.labels(endpoint='/predict/loan').observe(time.time() - start)
        REQUEST_COUNT.labels(method='POST', endpoint='/predict/loan', status=200).inc()
        return LoanPredictionResponse(approved=bool(prob >= 0.50), approval_probability=prob,
risk_tier=tier)
    except Exception as e:
        REQUEST_COUNT.labels(method='POST', endpoint='/predict/loan', status=500).inc()
        raise HTTPException(status_code=500, detail='Inference error: ' + str(e))

@app.post('/api/v1/predict/credit-limit', response_model=CreditLimitResponse)
async def predict_credit_limit(req: CreditLimitRequest):
    try:
        df_input = pd.DataFrame([req.model_dump()])
        pred = float(models['credit'].predict(df_input)[0])
        return CreditLimitResponse(recommended_credit_limit_inr=max(10000.0, round(pred, 2)),
currency='INR')
    except Exception as e:
        raise HTTPException(status_code=500, detail='Regression error: ' + str(e))
```

PO & PSO ATTAINMENT

PO / PSO No.	Graduate Attribute	Attained	Justification
PO 1	Engineering knowledge	Yes	Applied linear algebra, calculus, statistical probability, and machine learning principles to model complex retail credit default dynamics.
PO 2	Problem analysis	Yes	Formulated dual-objective credit decisioning mathematically, analyzing data leakage, class imbalance, and performance trade-offs.
PO 3	Design/development of solutions	Yes	Designed an end-to-end MLOps platform integrating Scikit-Learn pipelines, XGBoost estimators, and high-performance FastAPI microservices.
PO 4	Conduct investigations of complex problems	Yes	Conducted rigorous 5-fold cross-validation and empirical comparisons across 6 supervised baseline algorithms.
PO 5	Modern tool usage	Yes	Utilized modern tools: Python 3.12, XGBoost, DVC, MLflow, Docker, Prometheus, and Chrome DevTools.
PO 6	The engineer and society	Yes	Addressed consumer financial access, reducing underwriting latency from 72 hours to sub-15ms.
PO 7	Environment and sustainability	Yes	Optimized gradient tree boosting algorithms for fast training (0.34s), minimizing server compute consumption.
PO 8	Ethics	Yes	Enforced strict fair lending standards, auditability, data privacy, and mitigation of demographic biases.
PO 9	Individual and team work	Yes	Collaborated effectively across modular data engineering, backend APIs, and frontend client subsystems.
PO 10	Communication	Yes	Documented architectural diagrams, comprehensive report chapters, and published interactive Swagger documentation.
PO 11	Project management and finance	Yes	Managed multi-stage development milestones, tracked DVC data versions, and budgeted project deliverables.
PO 12	Life-long learning	Yes	Adapted evolving MLOps practices, containerization patterns, and regulatory guidelines to maintain technical relevance.
PSO 1	AI & Data Science System Design	Yes	Engineered and validated dual-model gradient boosting pipelines on 75,000 real-world financial records.
PSO 2	Enterprise Solutions & Deployment	Yes	Containerized microservice architecture delivering sub-15ms response times under 120 req/s load.

An MLOps-Driven Machine Learning Framework for Automated Loan Approval and Credit Limit Recommendation

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Abstract—Manual credit decisioning workflows in retail lending suffer from operational latency, human subjectivity, and data leakage. This paper presents an end-to-end Machine Learning Operations (MLOps) framework that automates binary loan approval and continuous credit limit recommendation. Utilizing Extreme Gradient Boosting (XGBoost) models coordinated with Scikit-Learn leakage-safe preprocessing pipelines, the classification model achieves a 0.9741 ROC-AUC, 89.00% accuracy, and 92.50% recall across 8,999 holdout applications. Concurrently, the regression model predicts credit card limits with an R^2 of 0.5141 and MAE of ₹192,670 INR. Experiments and artifacts are governed via MLflow, datasets are versioned via DVC, and predictions are served in real time via containerized FastAPI microservices monitored by Prometheus.

Keywords—Loan Approval, Credit Limit, MLOps, XGBoost, MLflow, FastAPI, DVC, Model Monitoring.

I. INTRODUCTION

Credit scoring has evolved from static linear scorecards into dynamic, automated machine learning pipelines. Lenders require systems that not only evaluate default risk instantaneously but also recommend prudent credit limits. Furthermore, modern regulatory standards mandate reproducible engineering: models must be trained without data leakage, tracked systematically, and monitored in production. This paper details the end-to-end design and empirical validation of such an MLOps-driven credit architecture.

II. RELATED WORK

Yeh and Lien (2009) established that nonlinear algorithms significantly outperform linear discriminant analysis in predicting default on credit card accounts. Chen and Guestrin (2016) demonstrated that XGBoost provides optimal regularization and computational efficiency for tabular records. Sculley et al. (2015) emphasized that production ML systems require extensive supporting infrastructure—encompassing schema validation, version control, and continuous testing—to prevent technical debt. Our work synthesizes these principles into an operational dual-model system.

III. METHODOLOGY & ARCHITECTURE

The proposed architecture operates across six decoupled layers. Raw data in data/raw/ is tracked with DVC hashes. Preprocessing pipelines fit SimpleImputer, StandardScaler, and OneHotEncoder exclusively on training partitions. Two XGBoost models are trained: a binary classifier evaluating loan default risk and a regressor predicting credit limit in INR (1 TWD = ₹2.99 INR). Both models are serialized as complete Scikit-Learn pipelines, tracked in an SQLite-backed MLflow registry, and deployed as asynchronous FastAPI microservices.

IV. EXPERIMENTAL EVALUATION

Experimental evaluation was conducted across 44,993 loan bureau applications and 29,944 credit profiles. On 8,999 unseen holdout applications, the XGBoost Classifier achieved an ROC-AUC of 0.9741, an accuracy of 89.00%, and a recall of 92.50%, outperforming Logistic Regression (ROC-AUC: 0.9522) and Random Forest (ROC-AUC: 0.9738). For credit limit recommendation, the XGBoost Regressor achieved an R^2 of 0.5141, an MAE of ₹192,670 INR, and an RMSE of ₹268,696 INR, surpassing Linear Regression (R^2: 0.4209). 5-fold cross-validation demonstrated high stability with standard deviations under 0.001.

V. SYSTEM DEPLOYMENT & TELEMETRY

The model serving layer is implemented in FastAPI and containerized using Docker. Prometheus scrapes invocation metrics and latency histograms via /metrics. The system delivers sub-15ms median response time under 120 req/s load. A client-side financial studio UI provides accessible multi-stage input guidance, debt-to-income derivation, and transparent post-decision credit options.

VI. CONCLUSION

The proposed MLOps platform bridges the gap between academic credit modeling and production financial engineering. By coupling high-recall XGBoost classification with calibrated continuous regression, the framework establishes an auditable, scalable blueprint for modern retail lending.